Lecture 18:

VLSI High-Speed I/O Equalization Circuits

Contents

This Topic: High-Speed I/O Equalization Circuits

Introduction

- Data rate of a typical high-speed I/O has been limited by two fundamental problems of the channel
  - Symbol Distortion
  - Time Dispersion
- If data rate is beyond the channel capacity, signal pulse cannot complete its transition within a symbol interval.
  - This appear as lack of zero crossings or inter-symbol interference (ISI)
- The equalizer is used to restore the signal timing and magnitude information

Introduction

- VLSI Equalizer can be classified in various ways
  - Based on its Structure:
    - Linear vs. Decision-feedback equalizers (DFEs)
  - Based on it location within I/O:
    - Tx vs. Rx equalizers
  - Based on circuit implementation
    - Analog vs. Digital equalizers
    - Continuous-Time vs. Discrete-Time
  - Based on coefficient updating
    - Fixed coefficient vs. Adaptive

Introduction

- Trade-offs between CTLE and DFE:
  – Linear equalizer can restore the original transmitted waveform, but it also amplifies high-frequency noise
  – Non-linear DFE can remove post-cursor ISI without amplifying high-frequency noise. However, it needs a clock

Channel Bandwidth Limitation

- I/O Channel bandwidth limitation is due to the frequency responses of the channel (usually lowpass)
- Key contributors to the channel attenuation:
  - Skin loss
  - Dielectric loss
- Other minor contributors including
  - Radiation of energy
  - Frequency response of the package parasitic
  - Lumped capacitance at the load/connector

Skin Loss Effect

- At high frequency (>100Mhz), current primarily flows on the surface of the conductor with skin depth of

$$\boxed{\delta = 1/\sqrt{\pi f \mu \sigma}}$$

- Resistance per unit length

$$R(f) = (1/2r)\sqrt{f\mu/\pi\sigma} \quad \text{(Round conductor of radius r)}$$

$$R(f) = (1/2W)\sqrt{\pi f\mu/\sigma} \quad \text{(Thin strip of width W)}$$

$$R(f) = K_s\sqrt{f}/d \quad \text{(General dimension d, Ks: constant)}$$

- Skin loss (attenuation) for length *l*

$$\boxed{A(f,l) = \exp(-\frac{R(f)}{Z_o}l)}$$

Dielectric Loss Effect

- Dielectric loss is the second major contributor to the channel

$$\boxed{R(f) = K_d f}$$

Constant

- Attenuation for a transmission line of length *l*

$$\boxed{A(f,l) = \exp(-\frac{R(f)}{Z_o} l)}$$

Typical Channel Frequency Responses

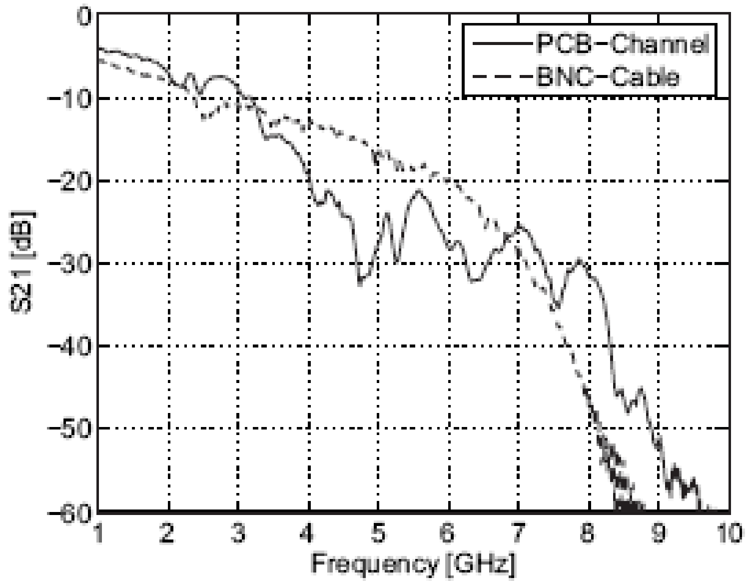
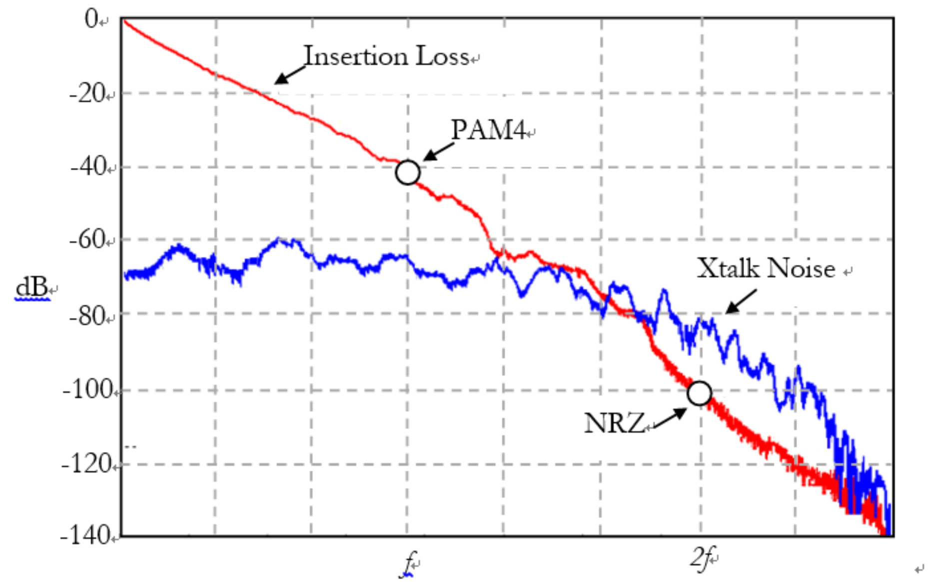
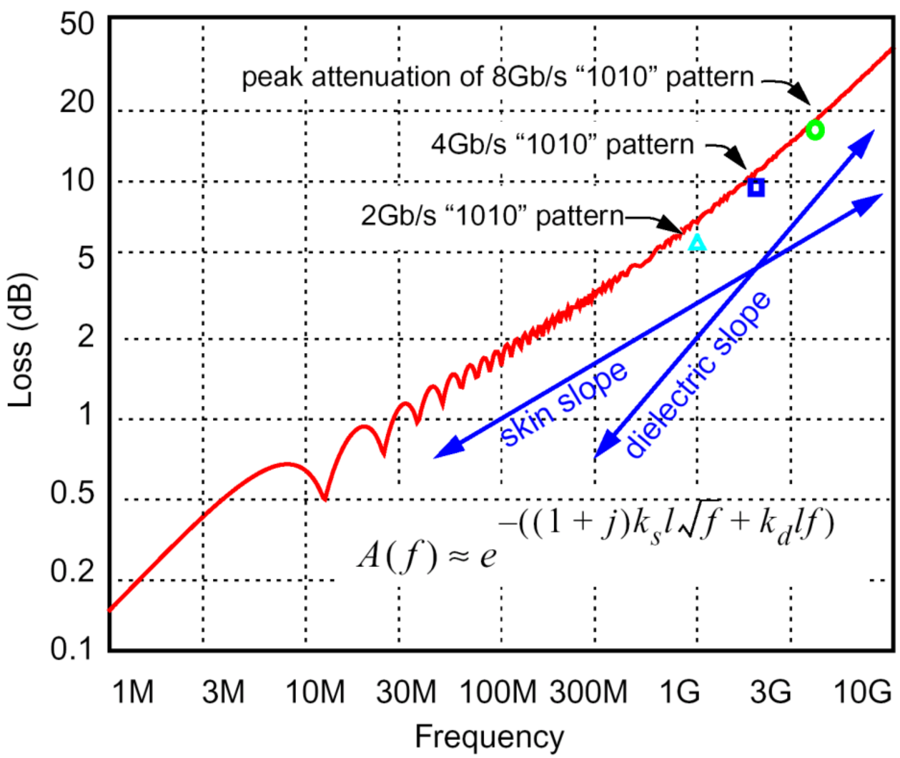


Fig. 9. Measured frequency response of the test channels. Each channel show high damping. Additionally the PCB-channel suffers from strong non-linear phase-distortion.



Skin and Dielectric Loss Model



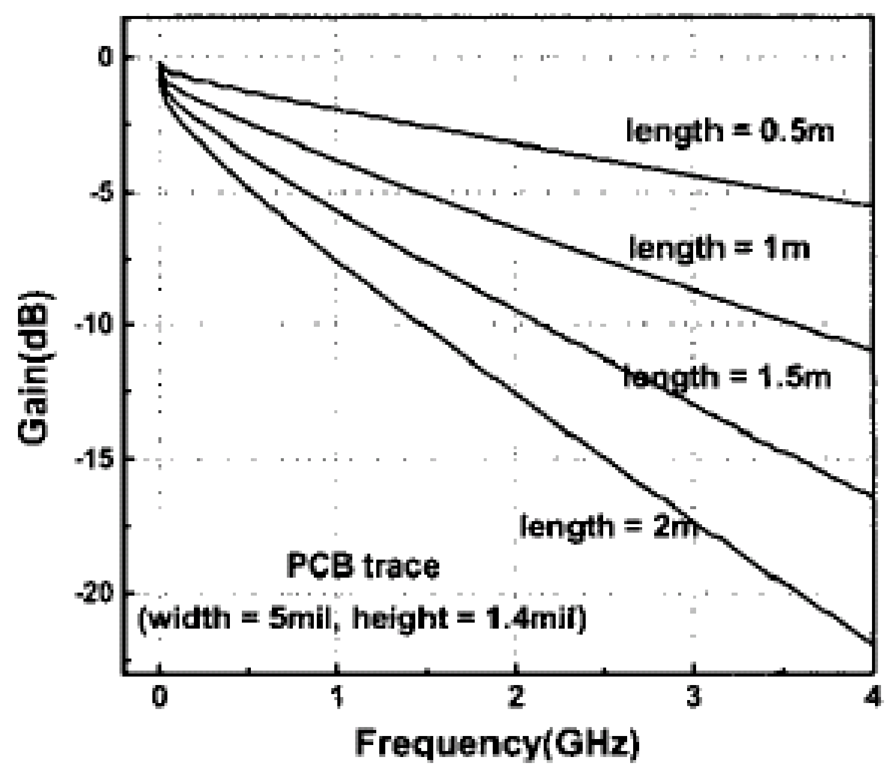
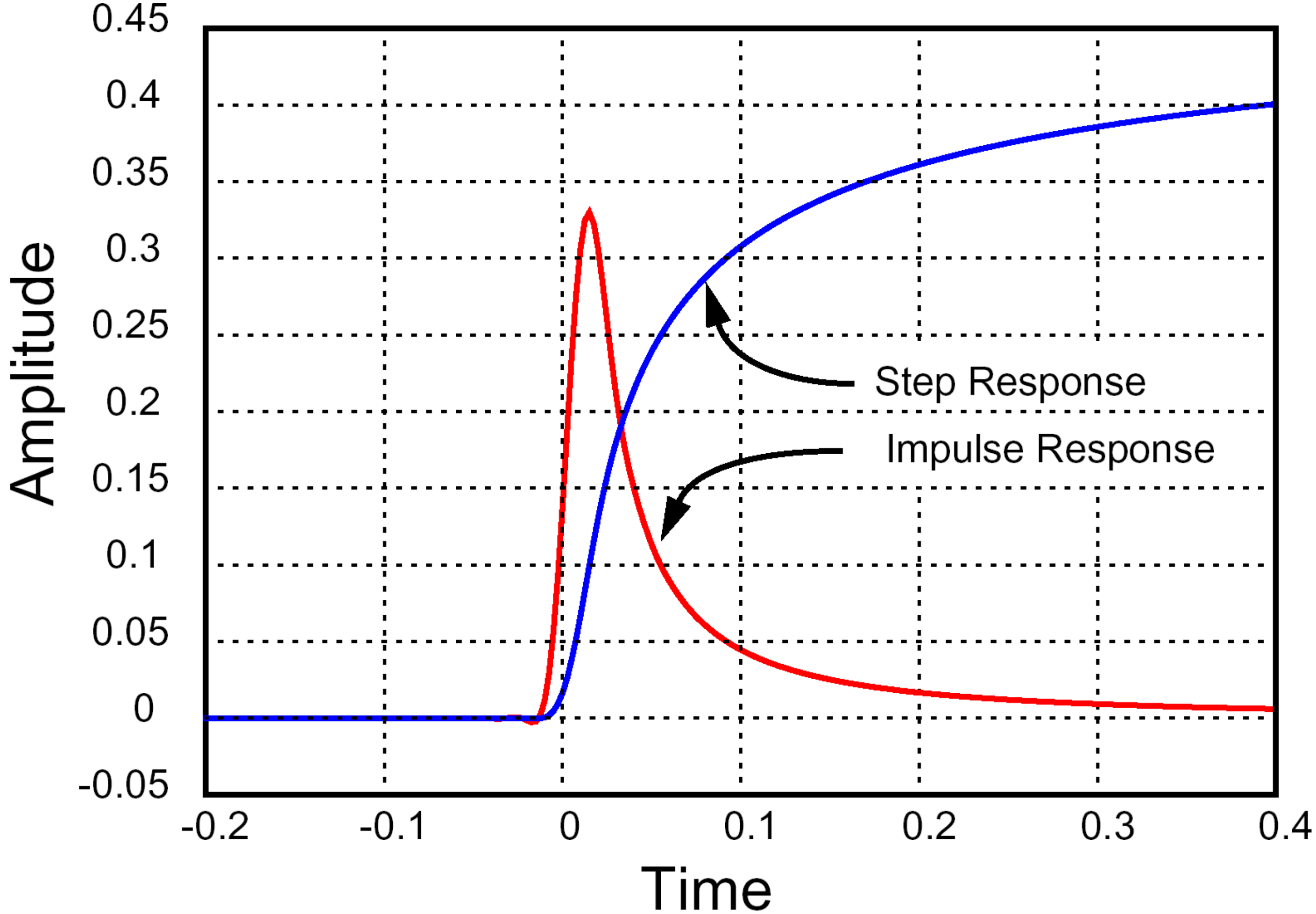


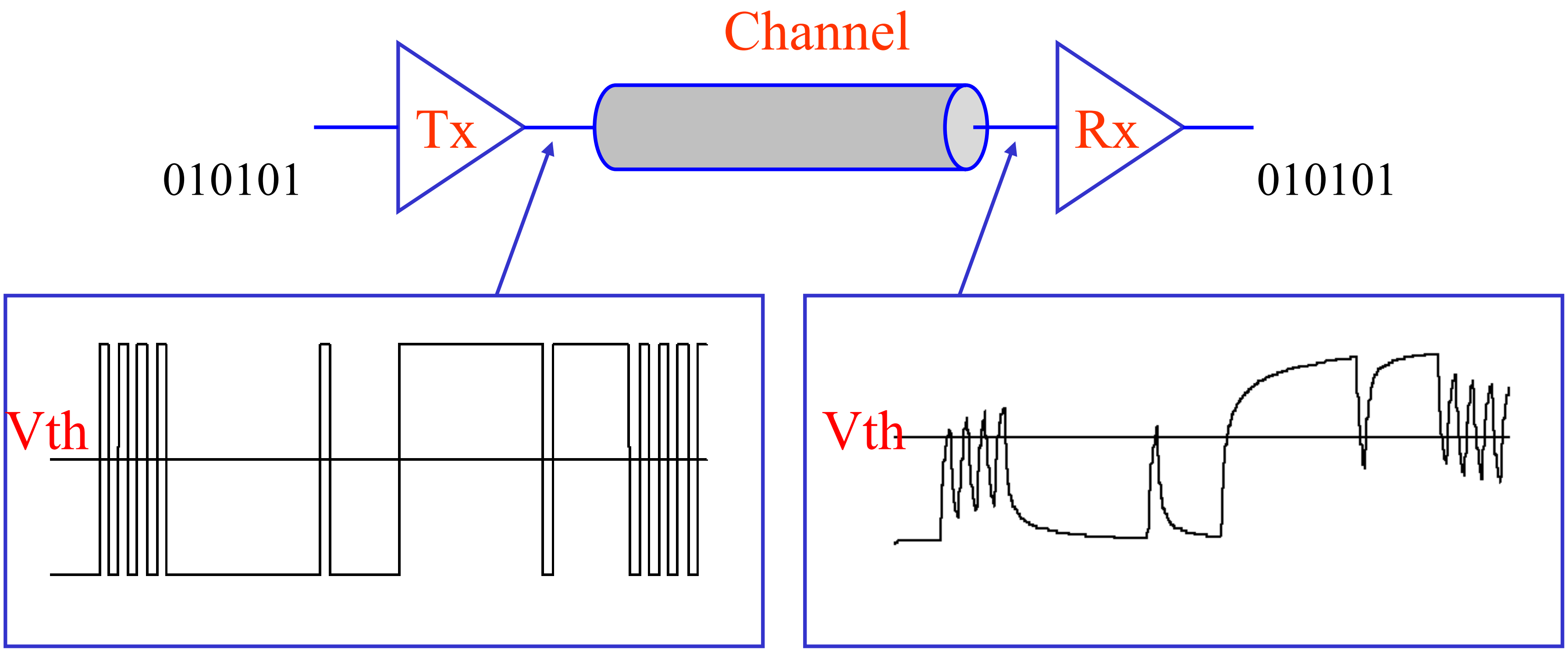
Figure 1. Attenuation curves for 50-ohm strip guided PCB traces.

Time Domain Response of Channel Bandwidth Limitation



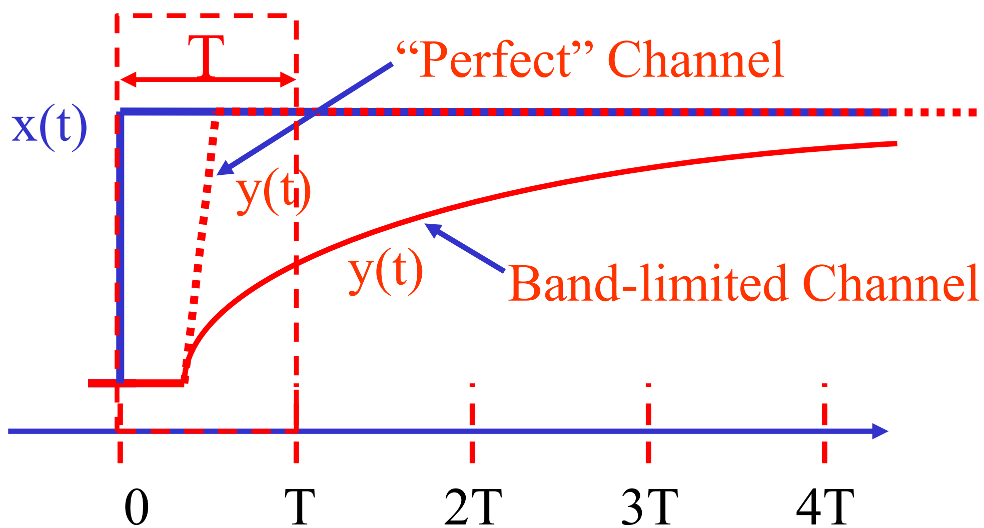
Effects of Channel Bandwidth Limitation

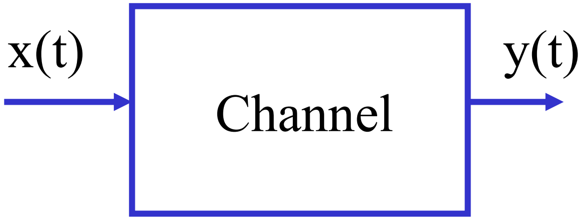
- High-speed signal in channel suffers from frequency dependent attenuation and which will cause intersymbol interference (ISI)



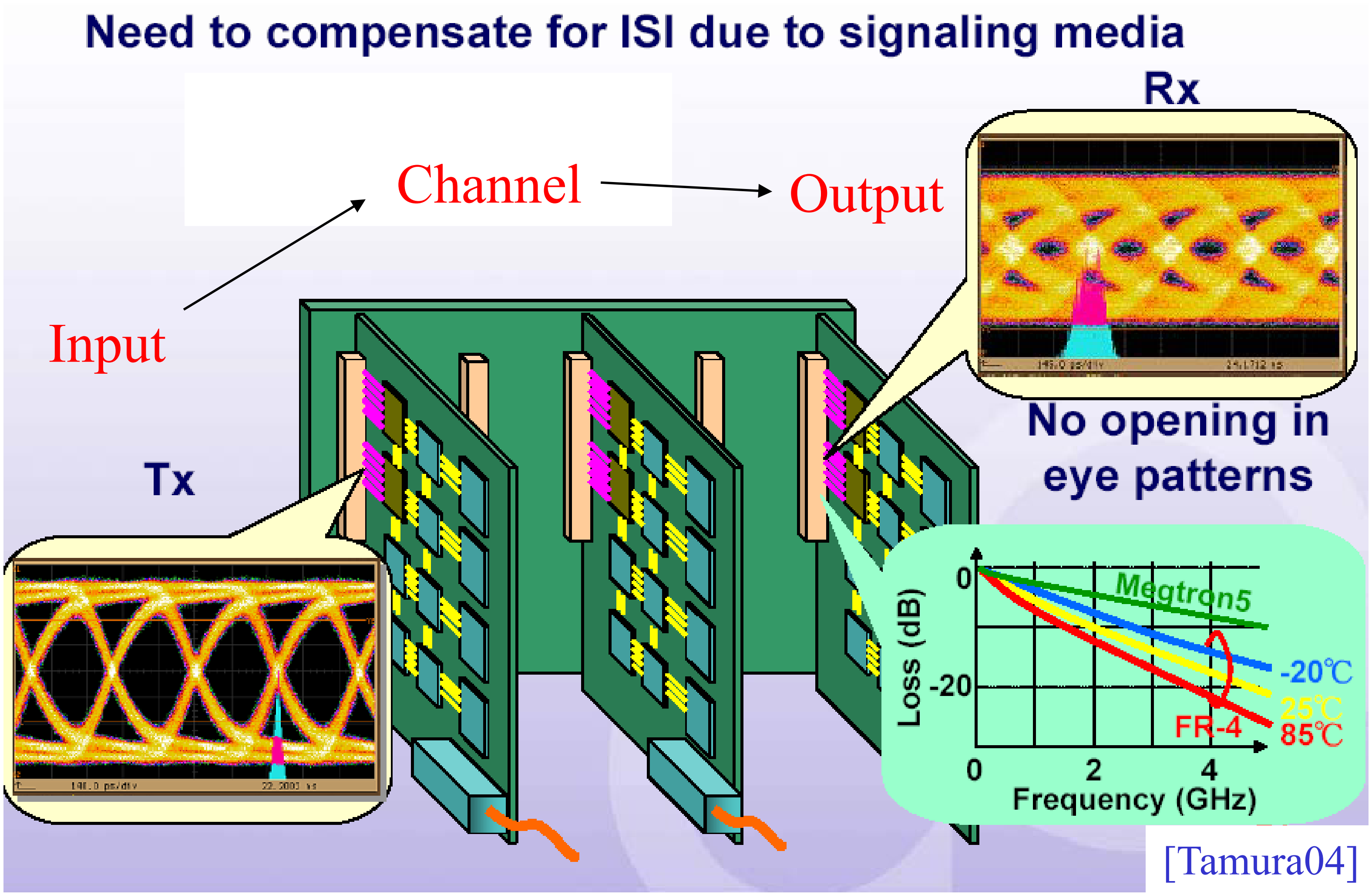
Inter-Symbol Interference (ISI) Model

- For binary data transmission, ISI is caused by the settling of data transition beyond the bit time
- It is resulted from band-limiting effects of the channel
- As the result, signal of current data bit is influenced by the previous data bits (pattern dependent)





Impact of ISI in High-Speed I/O Circuits

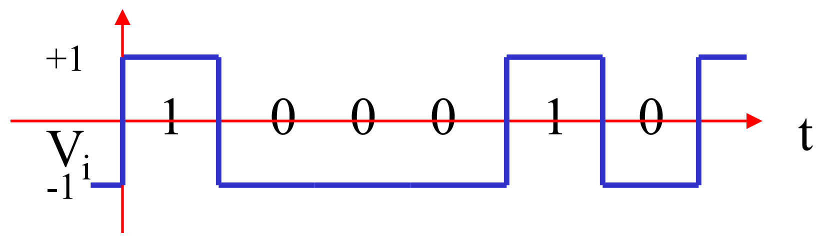


[Tamura04]

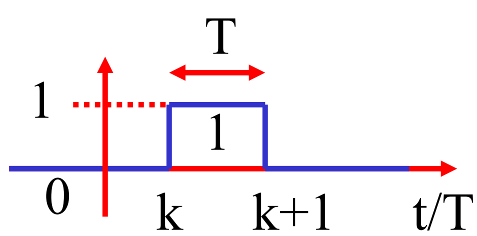
Digital Model of NRZ Data

- The NRZ data stream can be expanded into isolated pulse of “0”s and “1”s

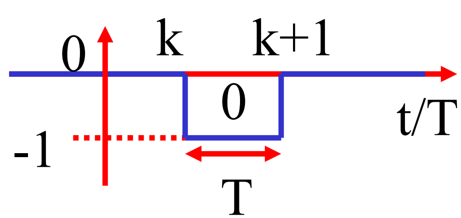
Data = “100010”



When Bit = “1”



When Bit = “0”

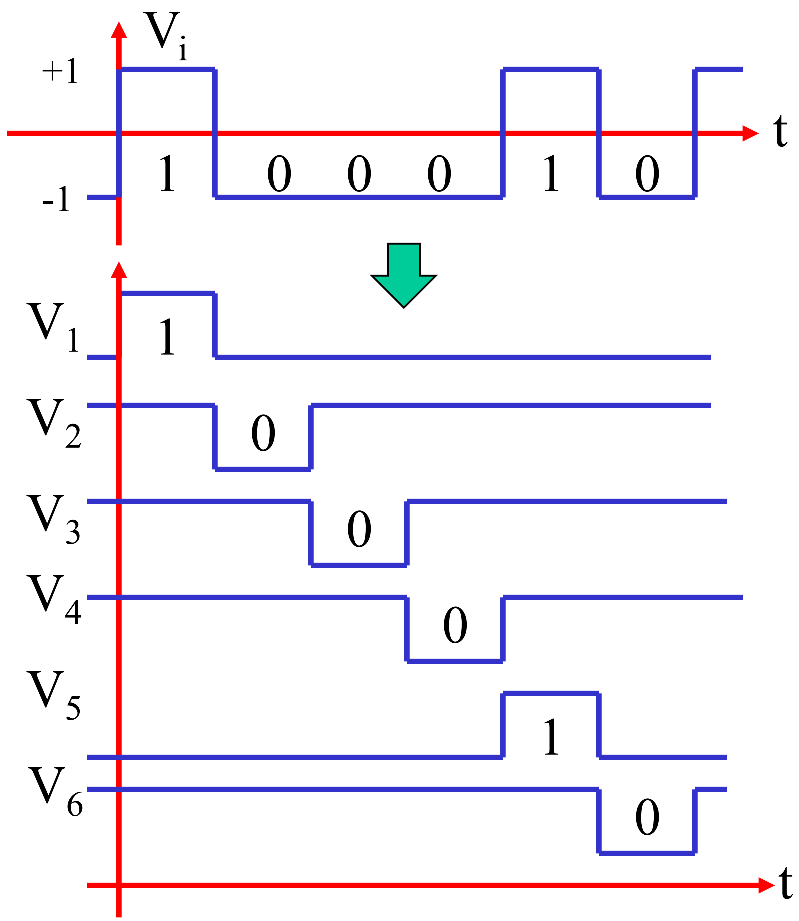


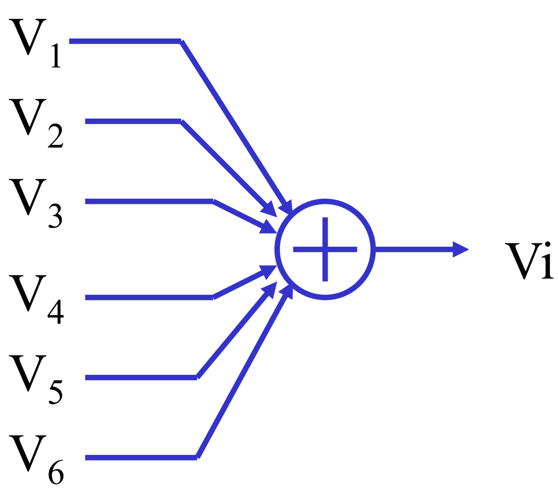
$$\begin{cases} P_k^{(1)}(t) \equiv h(t-kT) - h(t-(k+1)T) \\ P_k^{(0)}(t) = -P_k^{(1)}(t) \end{cases}$$

$$h(t) \equiv \begin{cases} 1 & t \geq 0 \\ 0 & t < 0 \end{cases}$$

Modeling of NRZ Data Stream

- Expansion of NRZ data stream using isolated pulses

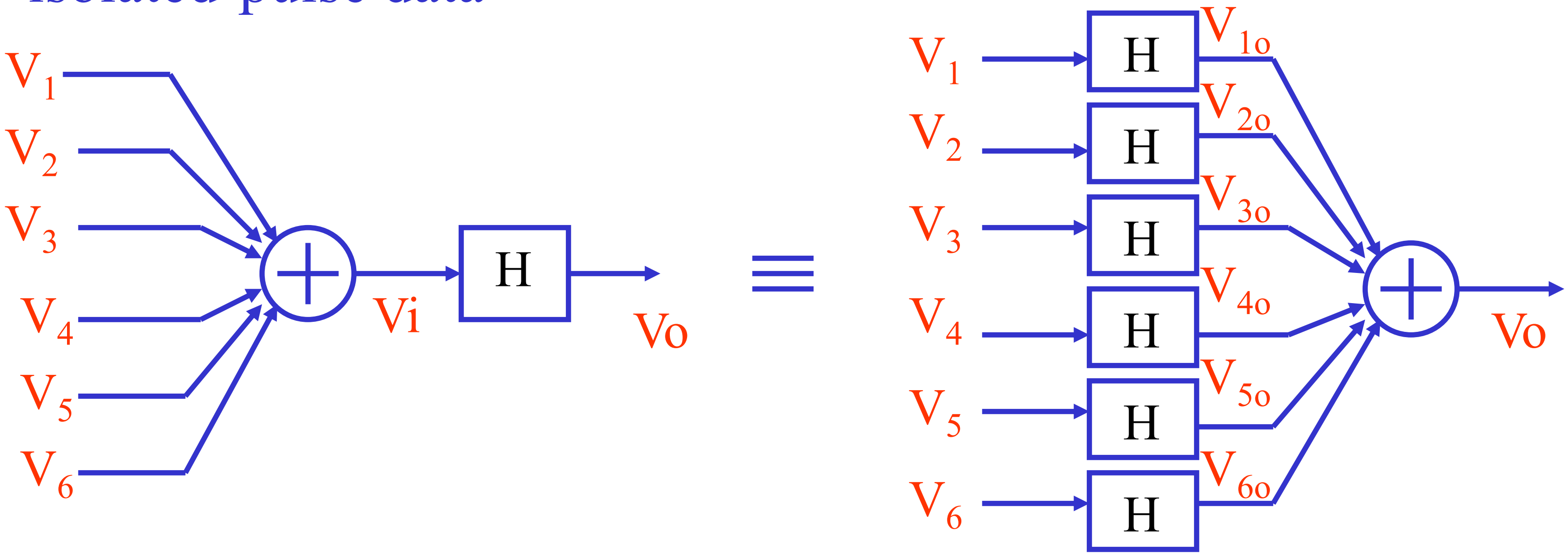




$$\boxed{V_i(t) = \sum_{k=-\infty}^{t} P_k^{(d_k)}(t)}$$

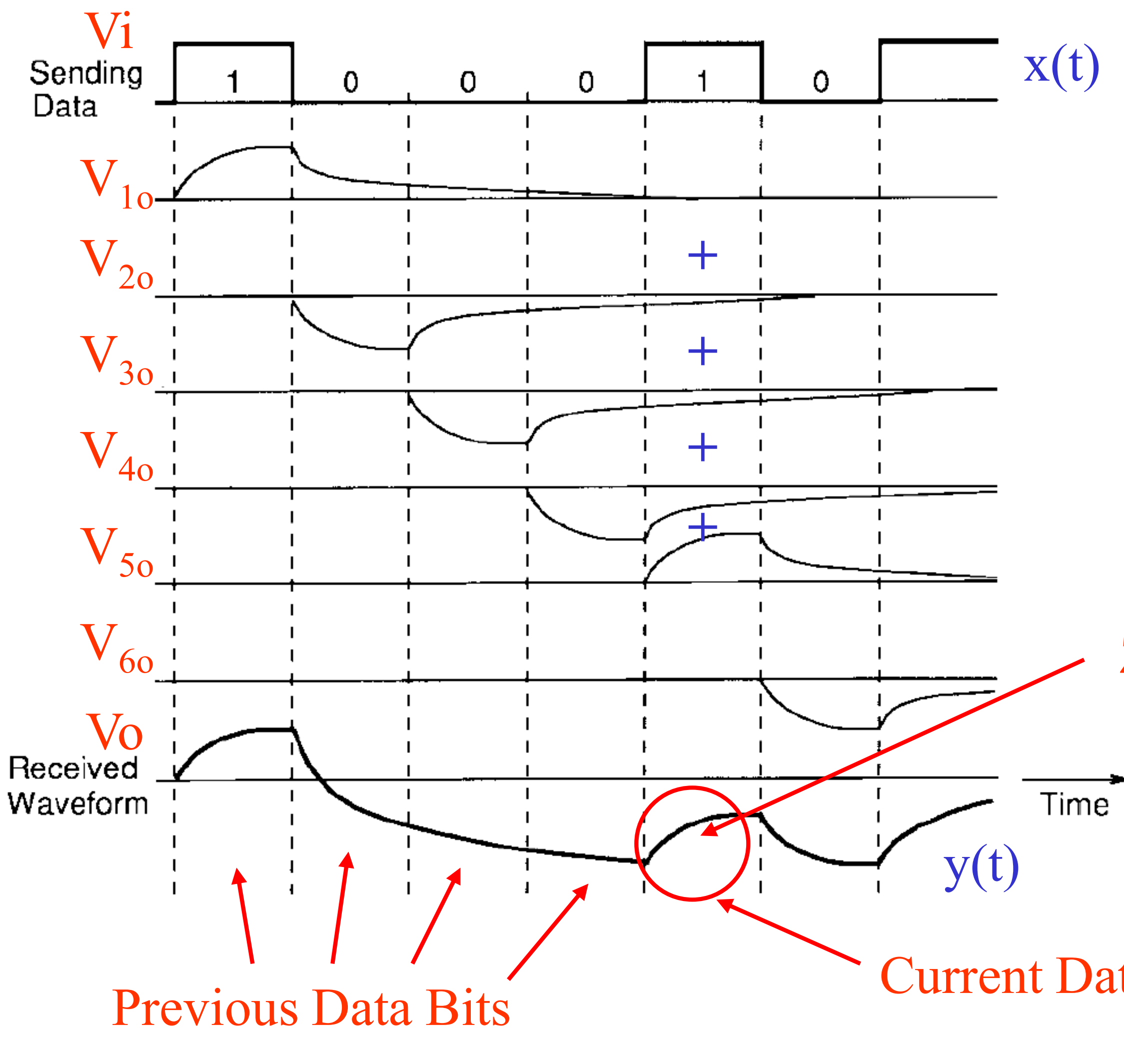
Channel Response to NRZ Data Stream

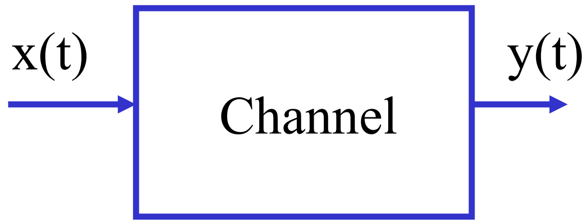
- For linear channel, the output is sum of channel to each isolated-pulse data



$$V_o(t) = H(V_i(t)) = \sum_{k=-\infty}^{t} H(P_k^{(d_k)}(t))$$

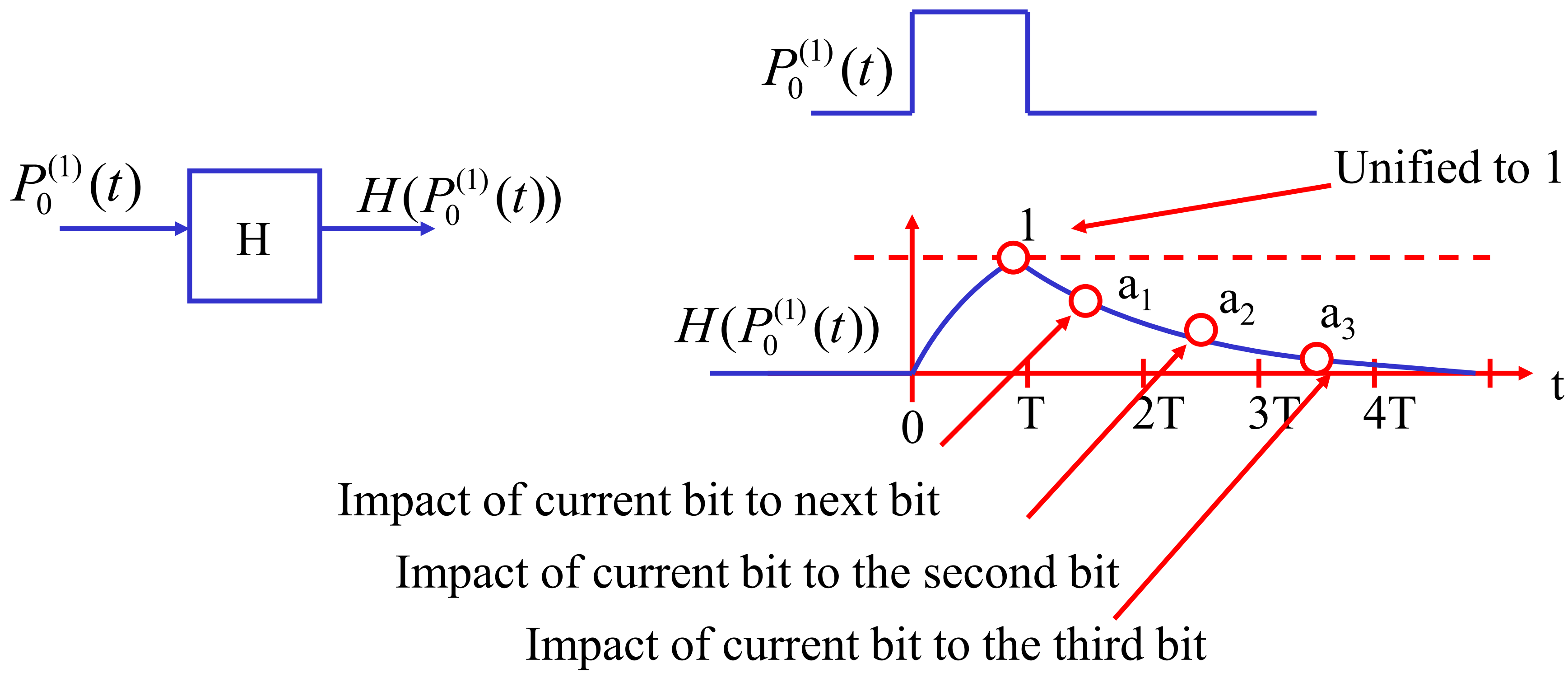
Inter-Symbol Interference (ISI)





Digitized Model of Channel Response

- The response of the channel can be characterized by its response to isolated pulse (e.g. Impulse response) as



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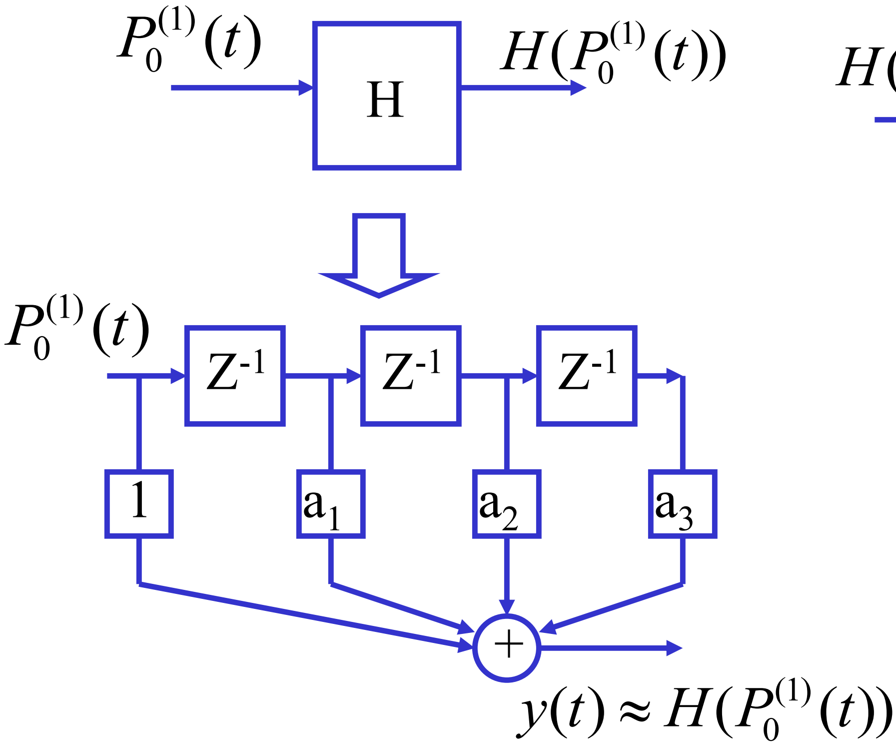
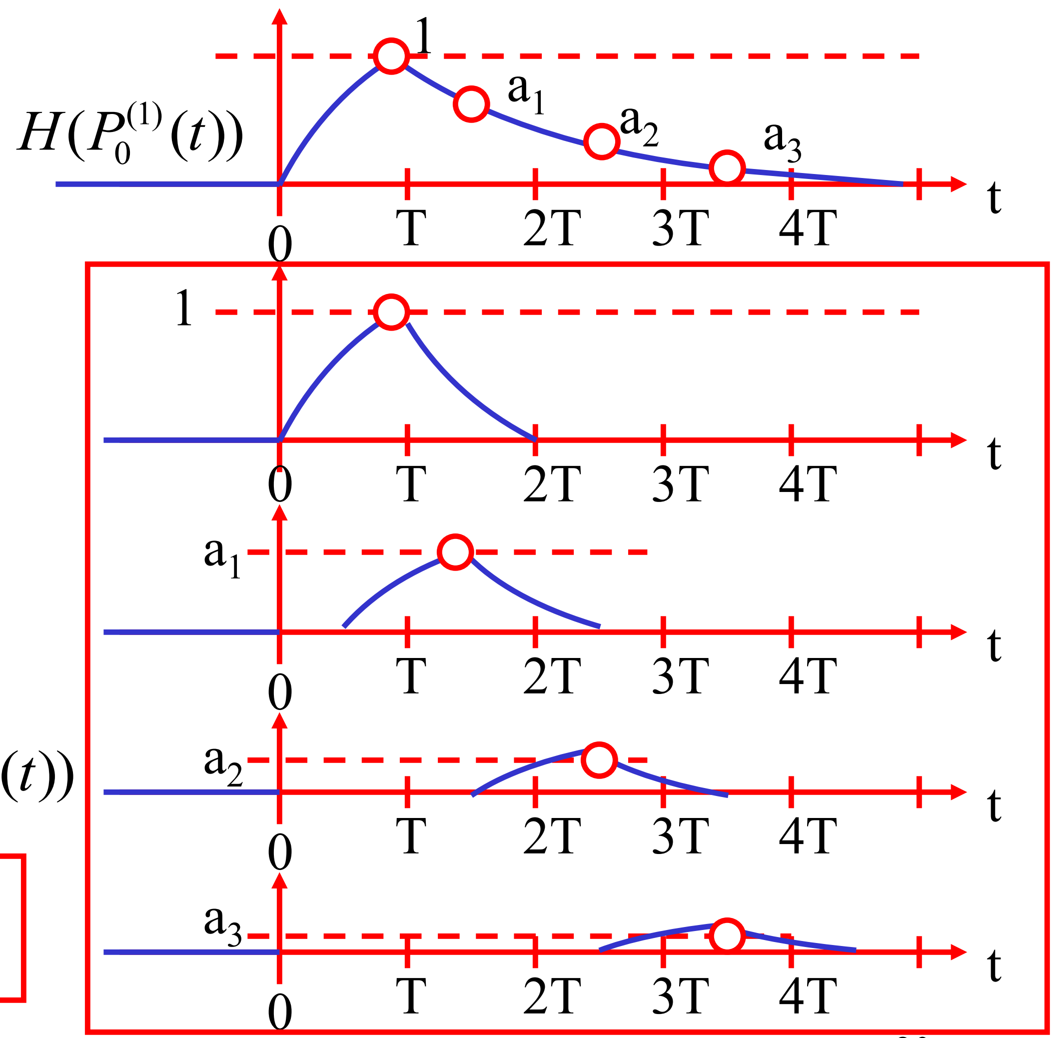
Digitized (FIR) Model of Channel





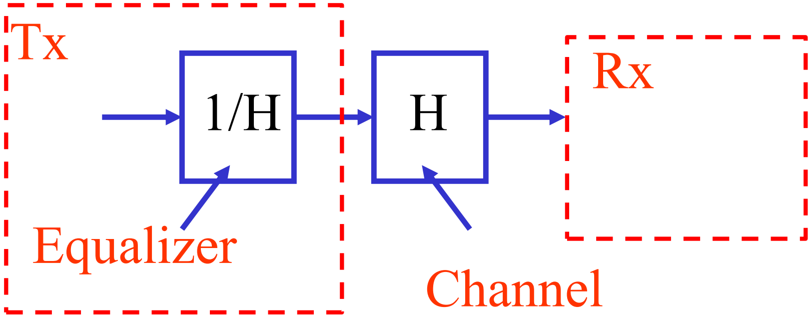
$$H \approx 1 + a_1 z^{-1} + a_2 z^{-2} + a_3 z^{-3}$$

VLSI Equalizer Circuit Implementations

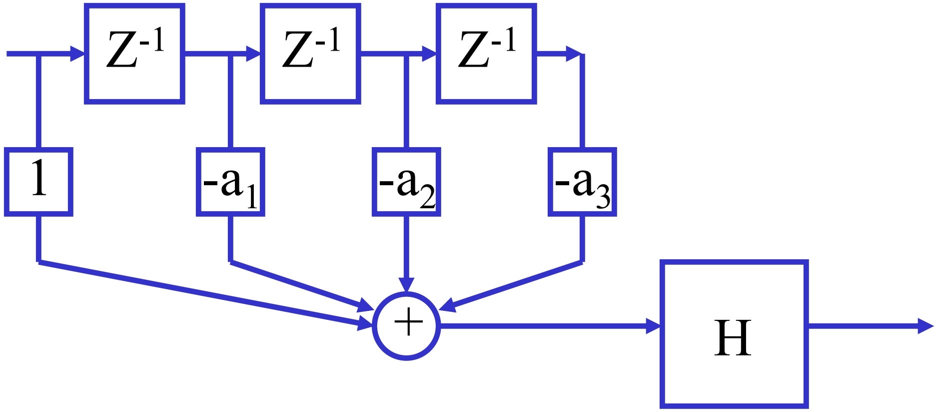
- Tx Equalizer
- DFE Equalizer
- Linear Equalizer
- TMX Equalizer
- SC Equalizer

Tx Equalizer

- Pre-emphasis or
- De-emphasis
- FIR filter



$$1/H \approx 1/(1 + a_1 z^{-1} + a_2 z^{-2} + a_3 z^{-3})$$
$$\approx 1 - a_1 z^{-1} - a_2 z^{-2} - a_3 z^{-3}$$



VLSI Tx FIR Equalizer Circuits

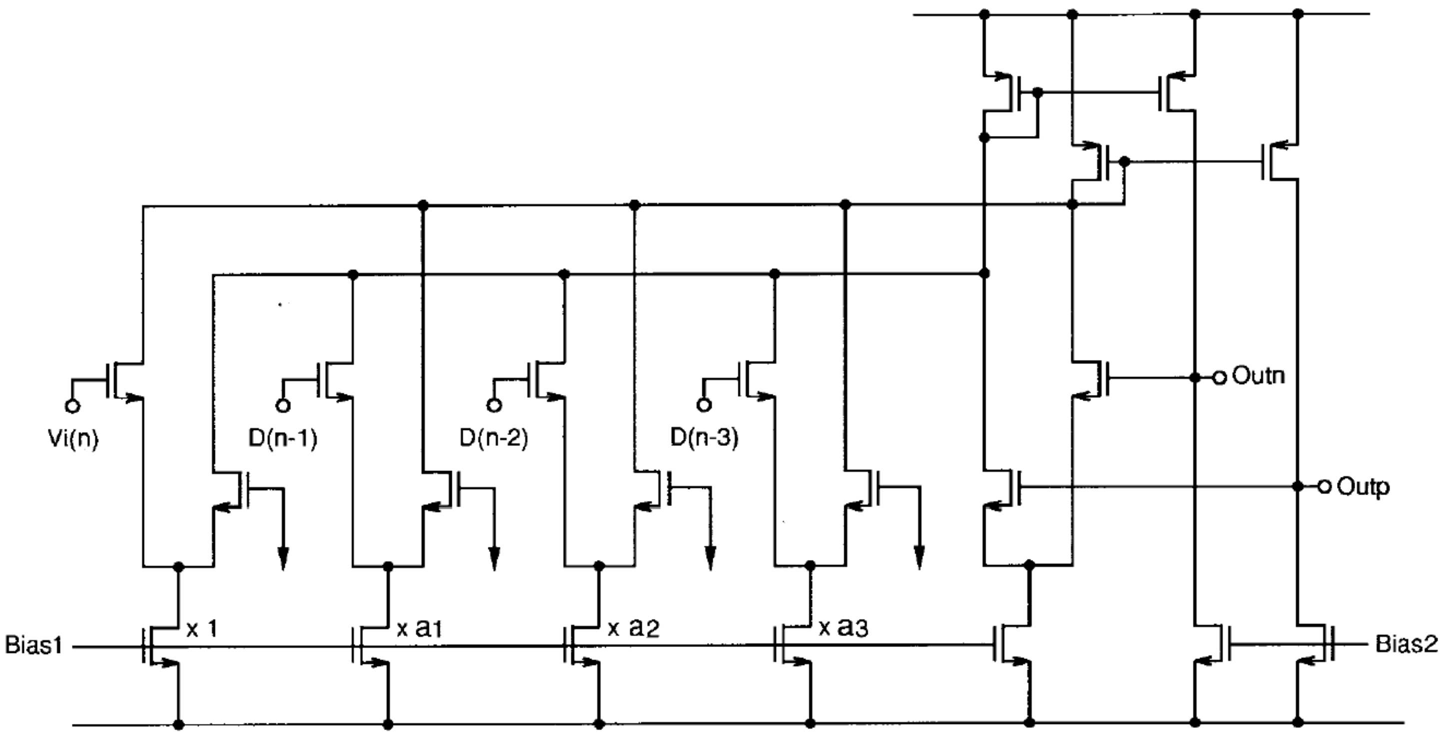
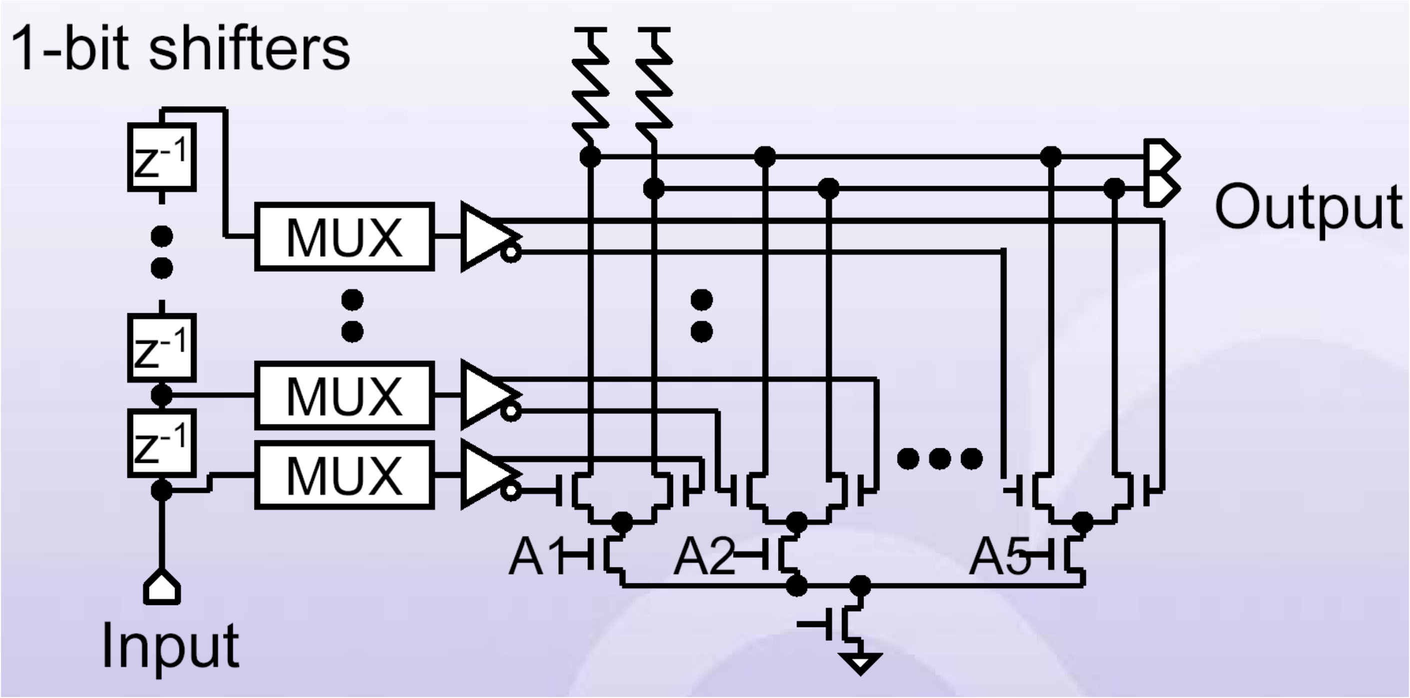


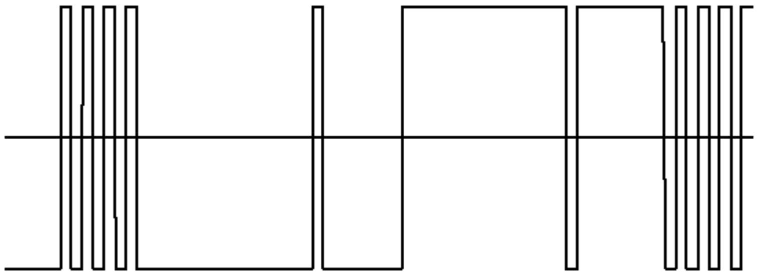
Fig. 8. Transversal filter in current domain.

VLSI Tx FIR Equalizer Circuits

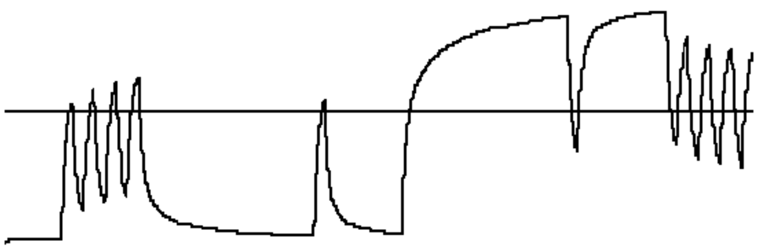


[Tamura04]

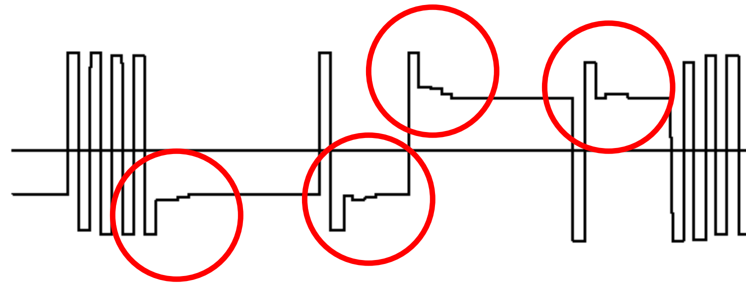
VLSI Tx FIR Equalizer Circuits



(a) Output of Tx w/o equalizer



(b) Input of Rx w/o equalizer

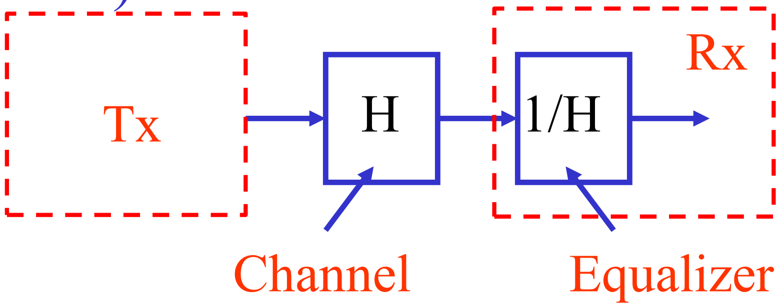


(c) Output of Tx w/ equalizer

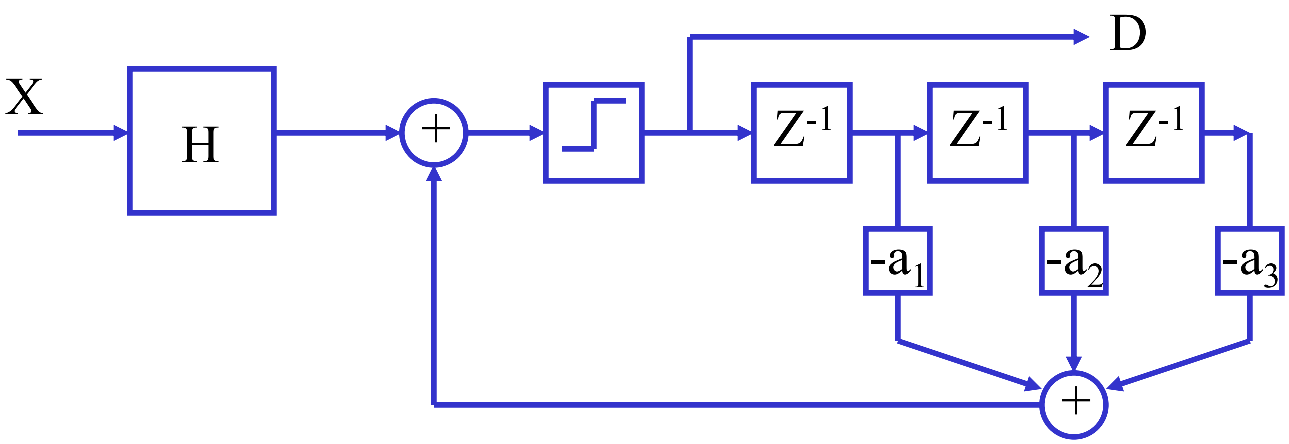
(c) Input of Rx w/ equalizer

Rx Equalizer

- Decision feedback equalization (DFE)
- IIR filter

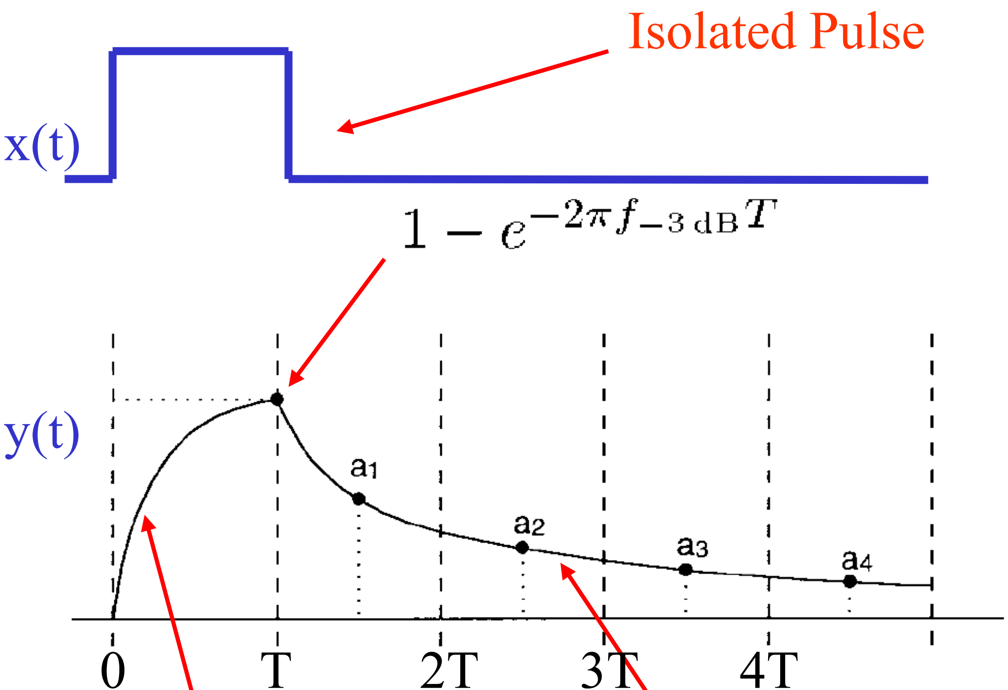


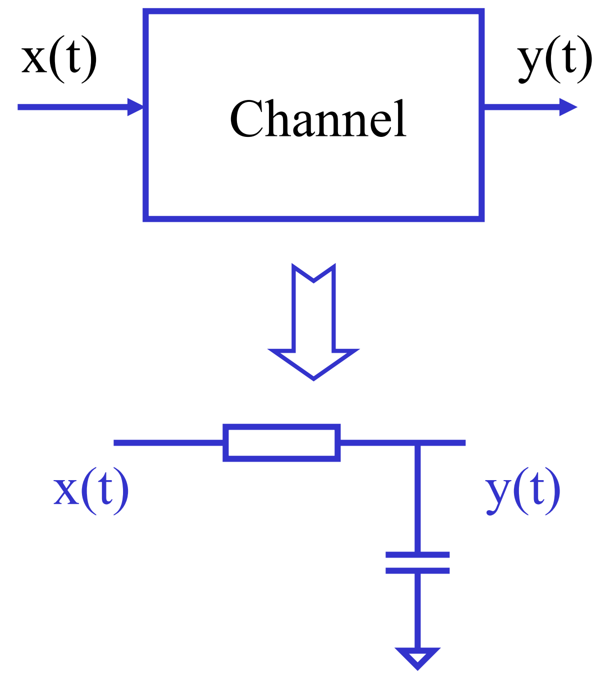
$$1/H \approx 1/(1 + a_1 z^{-1} + a_2 z^{-2} + a_3 z^{-3})$$



DFE Equalizer

- RC Approximation of the Channel Model





$$y(t) = 1 - e^{-2\pi f_{-3dB}\cdot t}$$

$$y(t) = (1 - e^{-2\pi f_{-3dB}\cdot T})e^{-2\pi f_{-3dB}\cdot (t-T)}$$

DFE Equalization

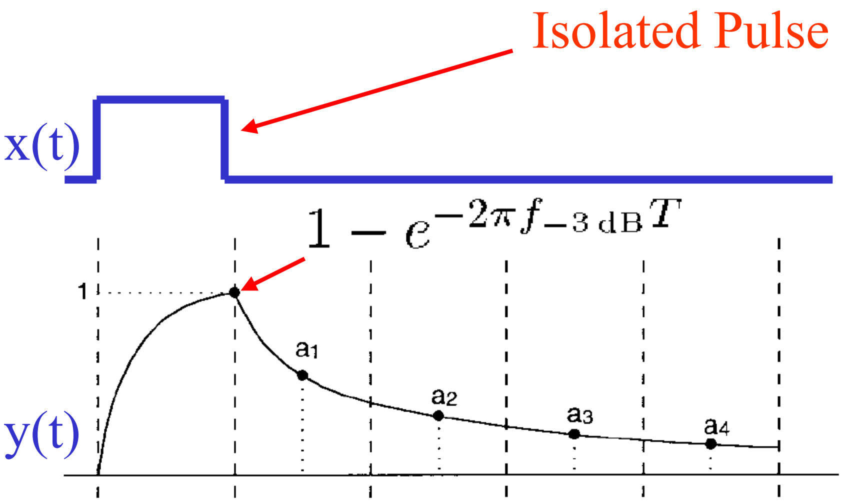
- RC Approximation of the Channel Model, the interference of neighboring bit is given by the coefficients as

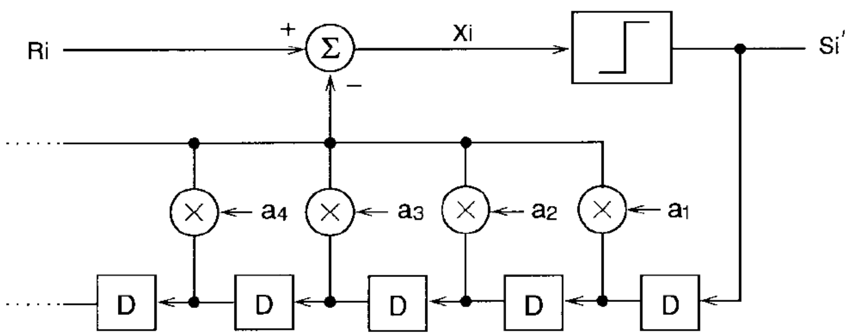
$$\boxed{a_k = (1 - e^{-2\pi f_{-3dB}\cdot T})e^{-2\pi f_{-3dB}\cdot (k-1/2)T}}$$

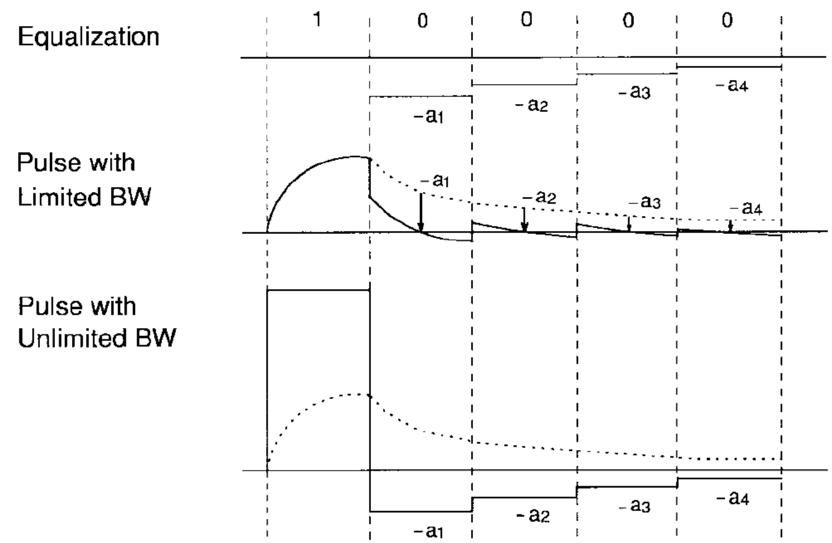
- That is the received data is a linear combination of current data bit and interference of previous data bits as

$$\boxed{R_i = x_i + a_1 \cdot x_{i-1} + a_2 \cdot x_{i-2} + a_3 \cdot x_{i-3} + \ldots}$$

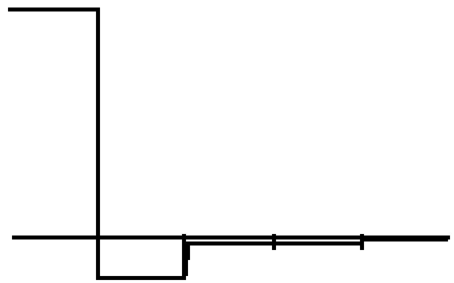
DFE Equalization



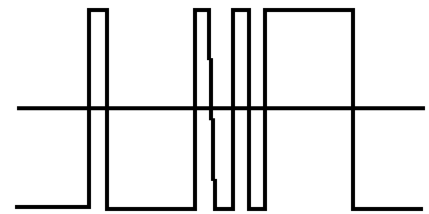




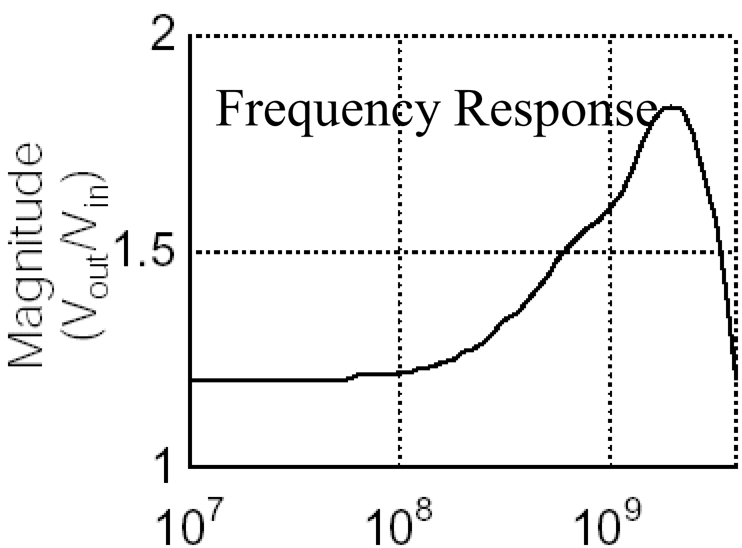
DFE Equalization



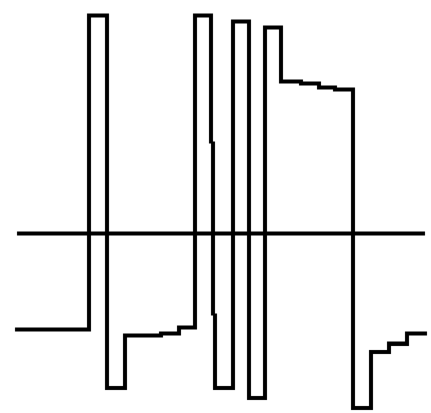
(a) Impulse Response



(c) Input of Equalizer



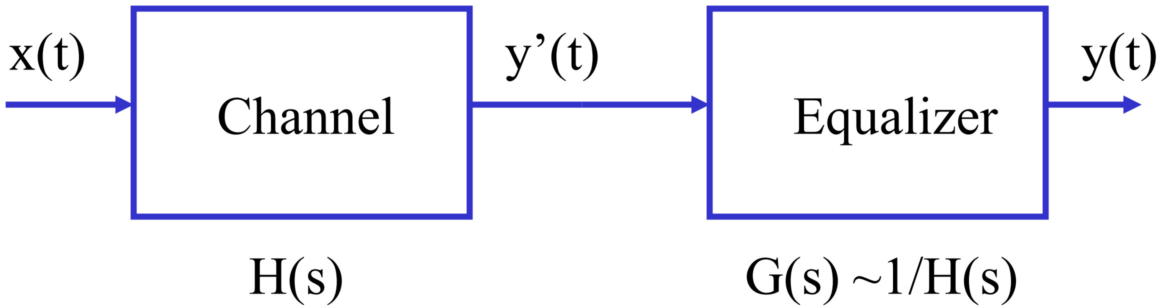
(b)

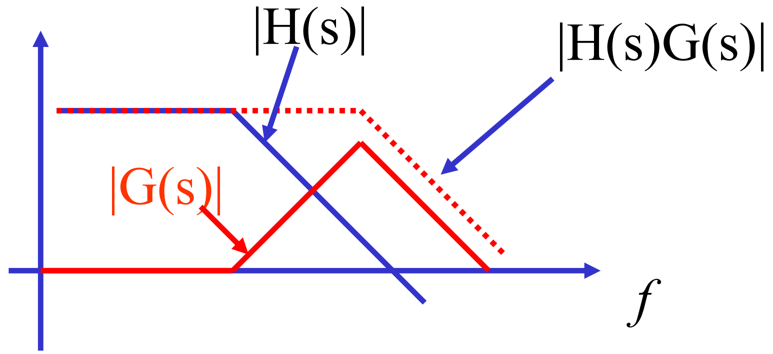


(d) Output of Equalizer

Linear Equalization

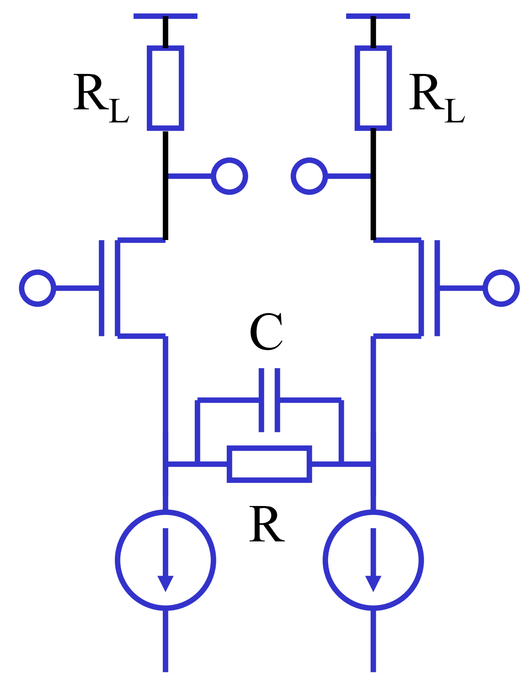
- Linear equalizers are based on a analog filter with inversed transfer function of the channel for the frequency of interests



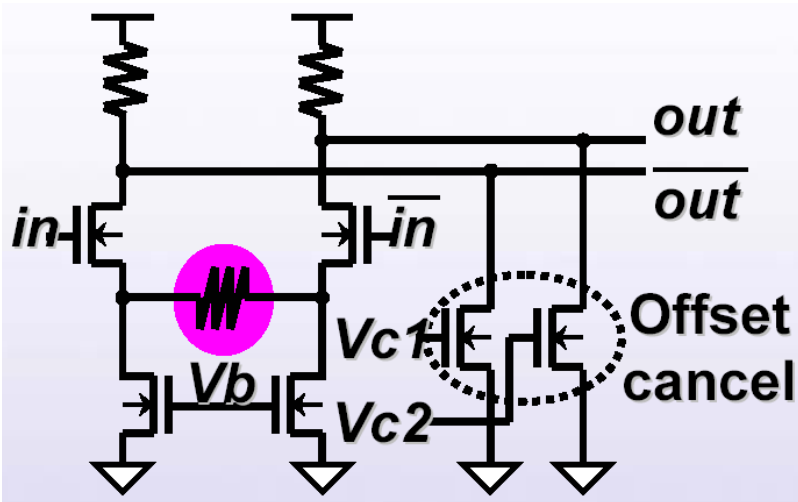


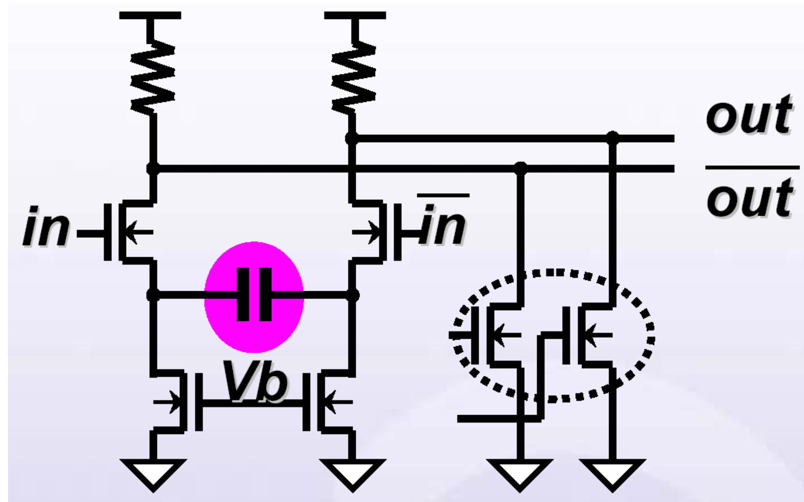
VLSI Linear Rx Equalizer Circuits

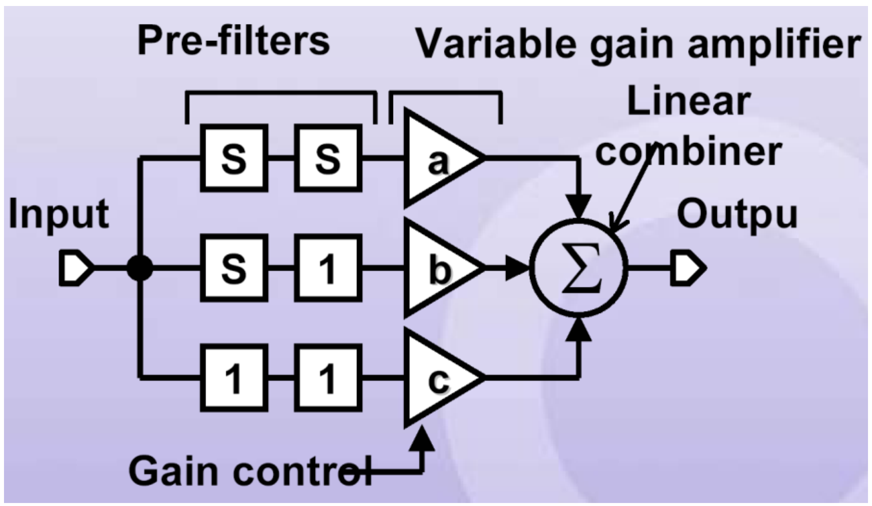
- High Frequency gain enhancement
  - Frequency dependent source de-generation



VLSI Linear Rx Equalizer Circuits







Transfer frunction :
$H(s) = as^2 + bs + c$

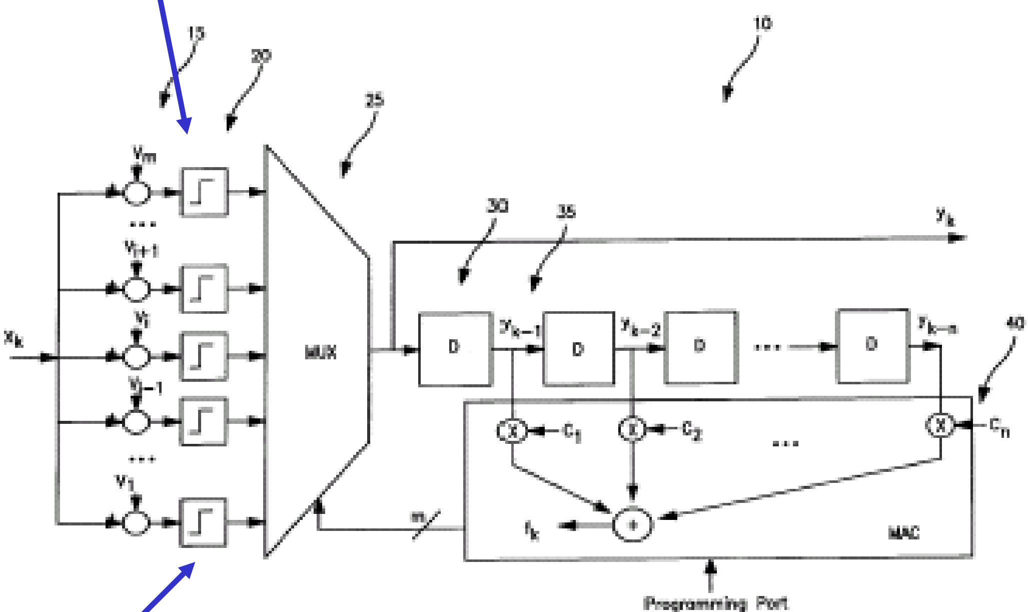
[Tamura04]

Threshold Multiplexing (TMX) Equalizer

- Parallel sampling

$$\frac{Y}{X}=\frac{1}{1+c_1z^{-1}+\ldots+c_nz^{-n}}$$

offset



sampler

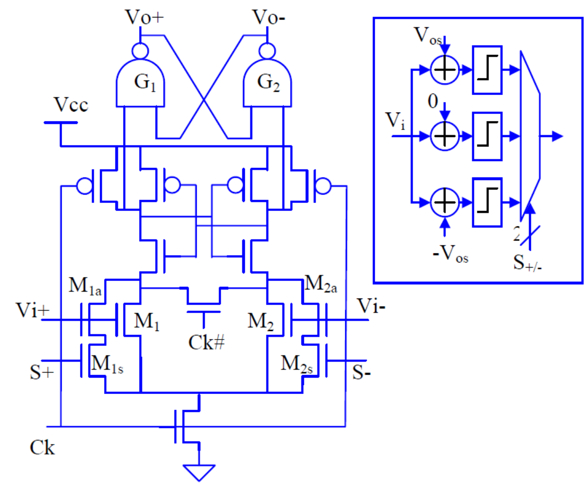
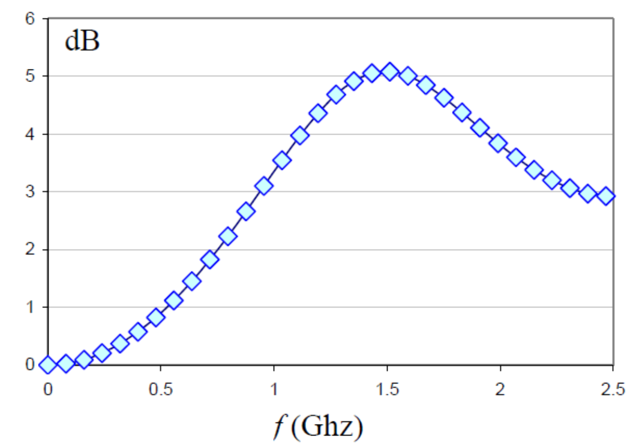
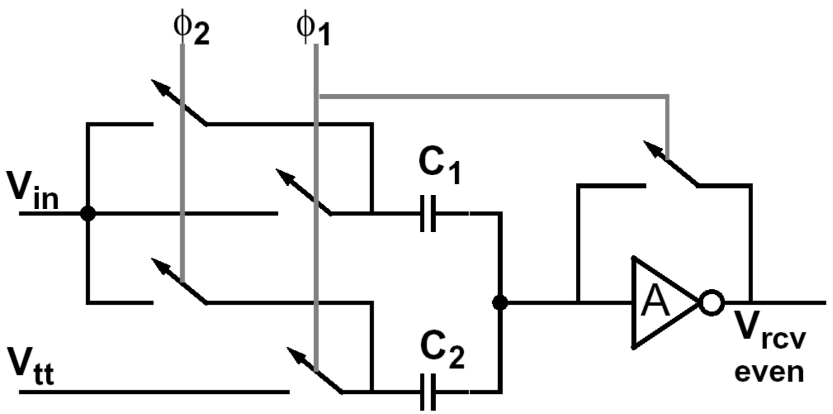


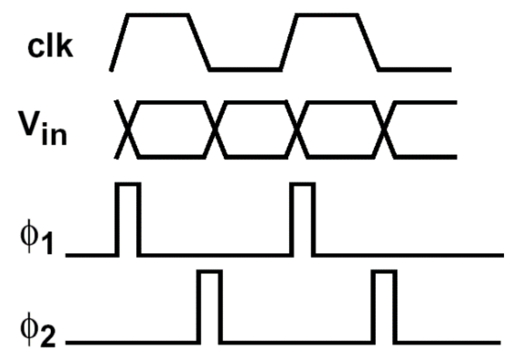
Figure 5: 3-level TMX sampler and equivalent circuit



SC DTLE Rx Equalizer Circuits

- $\Phi 1$ phase
  - V(n-1) stored on C1
  - C2 and amplifier resets
- $\Phi 2$ phase
  - V(n) stored on C2
  - V(n)-V(n-1) stored on C1
- Vrcvd = b(V(n) - aV(n-1))
- α and β depend on capacitances and parasitics





Reading Assignment

- Please read Chapter 13 of the textbook

VLSI High-Speed I/O Circuits

Theoretical Basis, Architecture, Modeling and Circuit Implementation

VLSI High-Speed I/O Circuits
By: Hongjiang Song

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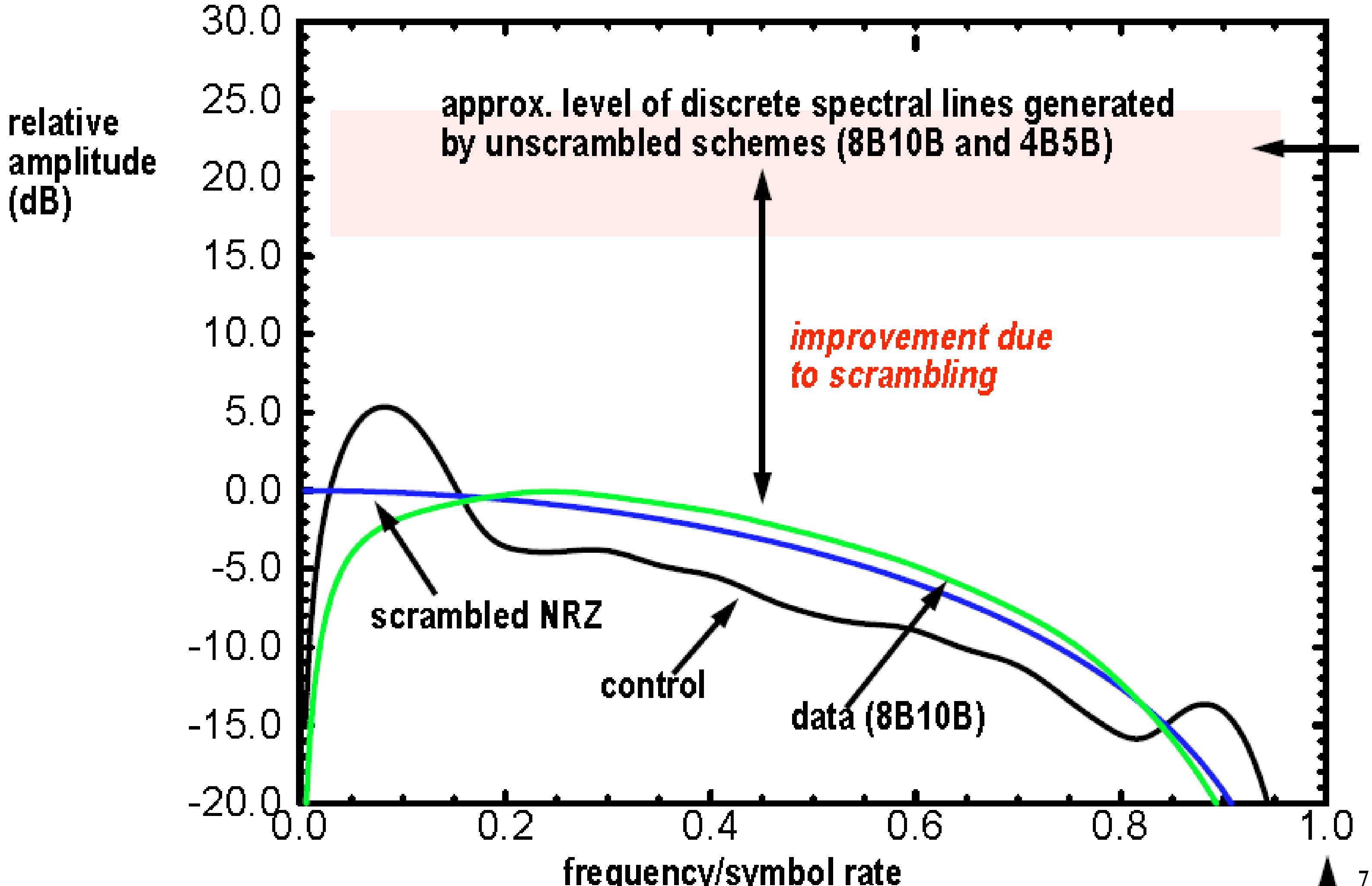
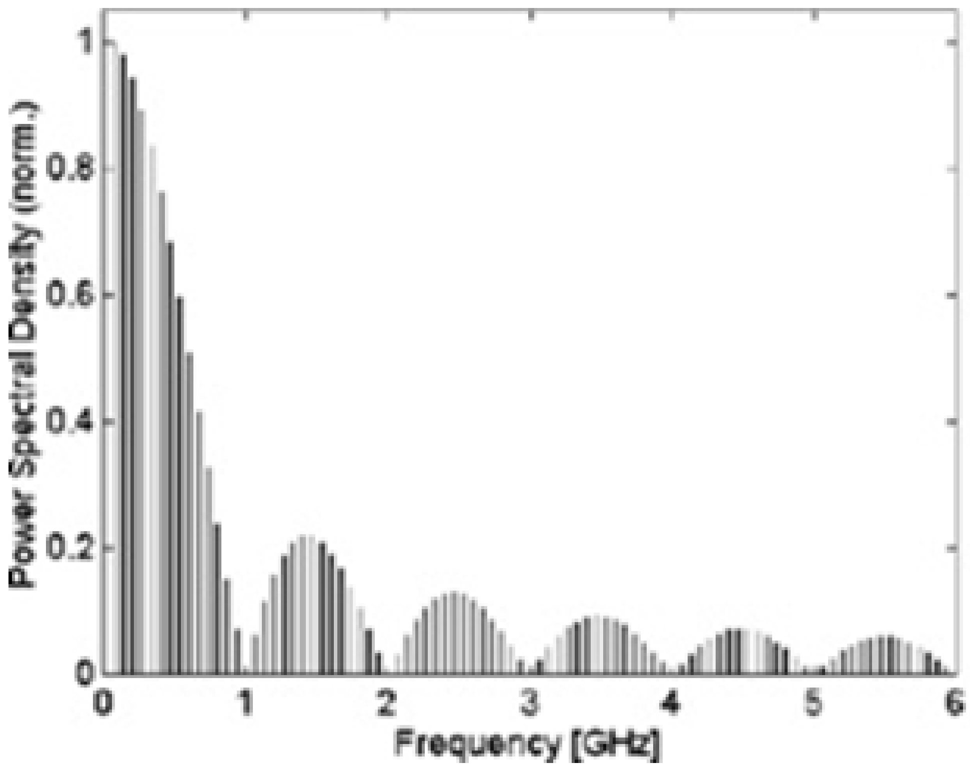
Power Spectrum of Data Formats



- Backup Foils



Case Study: MOS-C Equalizer

FULLY INTEGRATED MOSFET–C VARIABLE EQUALIZER CIRCUIT WITH ON–CHIP AUTOMATIC TUNING

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Abstract

A MOSFET–C variable bump equalizer with on–chip automatic tuning circuit was developed and fabricated in a $2\mu m$ CMOS process. The center frequency (ω_O), the gain at ω_O (G), and the 3db bandwidth (BW) of the bump equalizer are independently programmed by external DC control voltages. A SC gain control loop is used to provide on–chip automatic tuning.

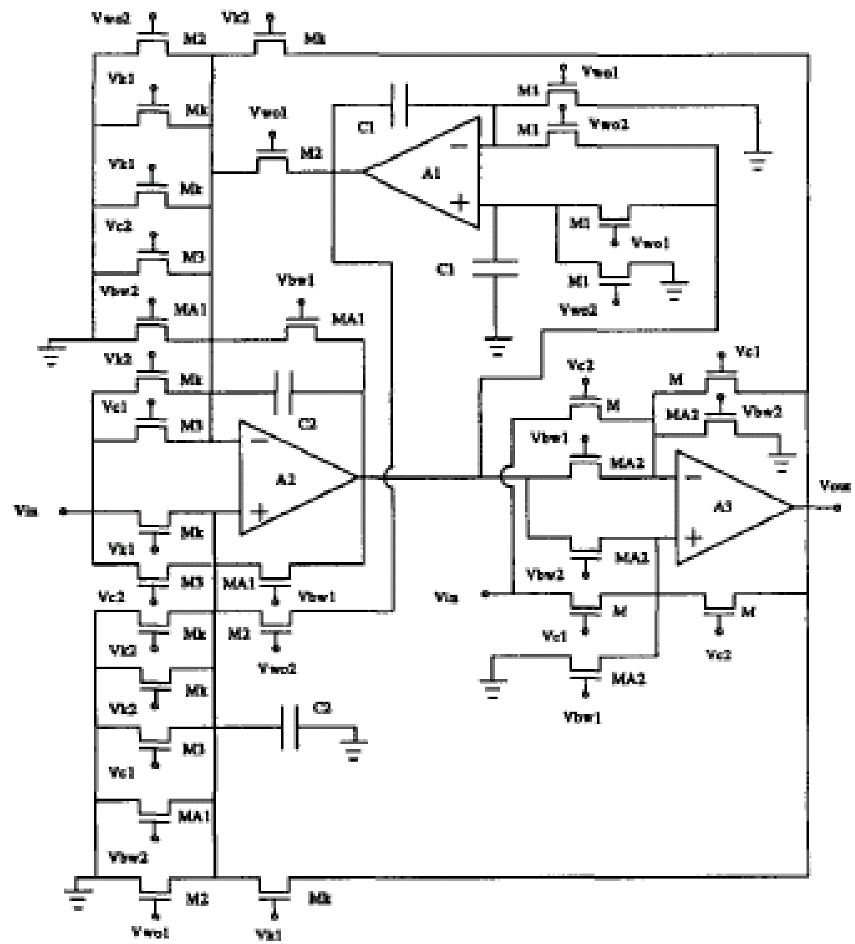

Figure 1: MOSFET-C bump equalizer

Case Study: Adaptive Equalizer

A 2-Gbps CMOS Adaptive Line Equalizer

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ABSTRACT

A 2-Gbps line equalizer circuit is realized with 0.25 ㎛ CMOS technology. The equalizer is made of input stage buffer, limiter and square difference circuits. The limiter has replica-feedback limiting amplifiers, which do not require common mode feedback. Successful equalization is demonstrated for signals transmitted over 1.5m long PCB trace.

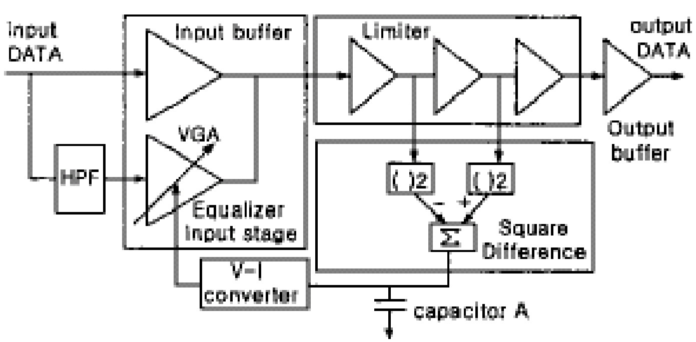


Figure 3. Equalizer block diagram.

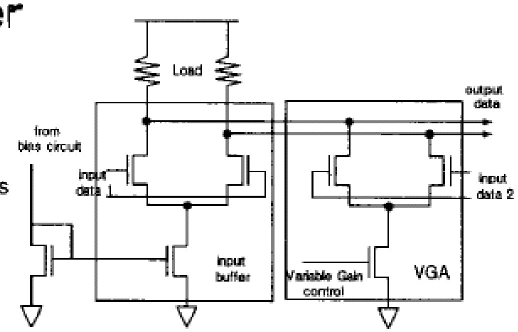


Figure 5. Equalizer input stage schematic

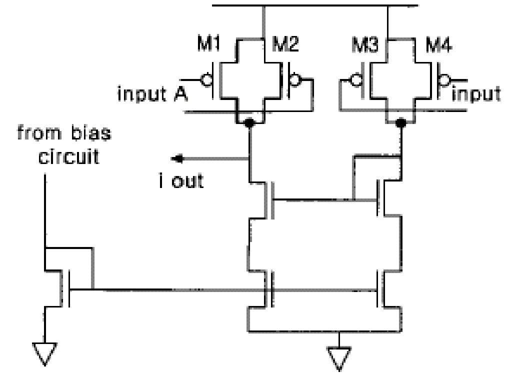


Figure 7. Square difference schematic

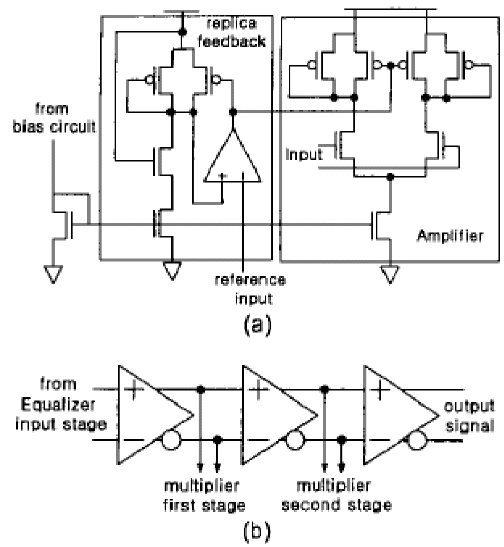


Figure 6. Limiter circuit: (a) Replica-feedback amplifier schematic, (b) Limiter schematic

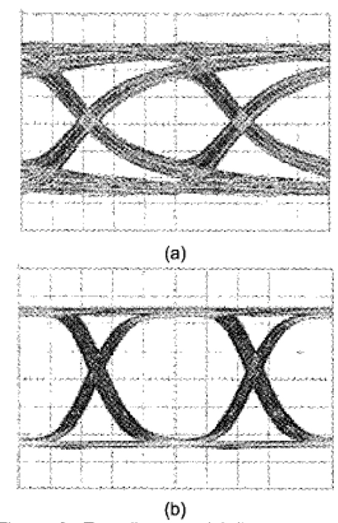


Figure 9. Eye diagram; (a) line output after 0.6m PCB trace, (b) equalizer output after 0.6m PCB trace. Vertical scale is 100mV/div; horizontal scale is 100ps/div.

2004 IEEE Asia-Pacific Conference on Advanced System Integrated Circuits(AP-ASIC2004)/ Aug. 4-5, 2004

Case Study: DDJ Equalizer

IEEE 2004 CUSTOM INTEGRATED CIRCUITS CONFERENCE

A 10Gb/s Data-Dependent Jitter Equalizer

James Buckwalter and Ali Hajimiri

California Institute of Technology
Pasadena, CA 91125 USA

Abstract

An equalization circuit is presented that reduces data-dependent jitter by aligning data transition deviations. This paper presents an analytic solution to data-dependent jitter and demonstrates its impact on the phase noise of the recovered clock. A data-dependent jitter equalizer is presented that compensates the impairment of the channel and lowers the phase noise of the recovered clock. The circuit is implemented in a SiGe BiCMOS process and operates at 10Gb/s. It suppresses phase noise resulting from data-dependent jitter by 10 dB.

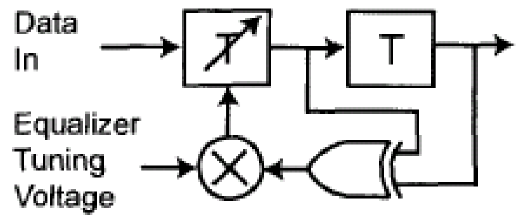


Fig. 4 The block diagram for a DDJ equalizer.

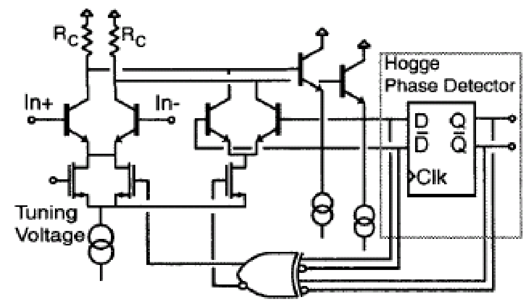


Fig. 6 Schematic for fully differential DDJ equalizer built into Hogge phase detector.

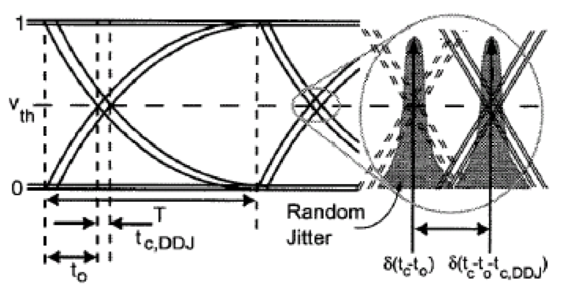


Fig. 1 The data-dependent jitter apparent in the data eye and the resulting jitter

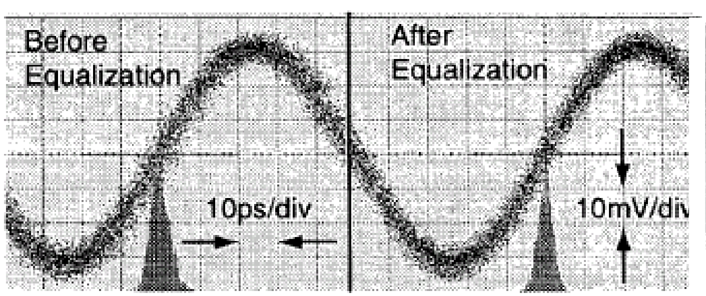


Fig. 8 Clock jitter with and without equalization.

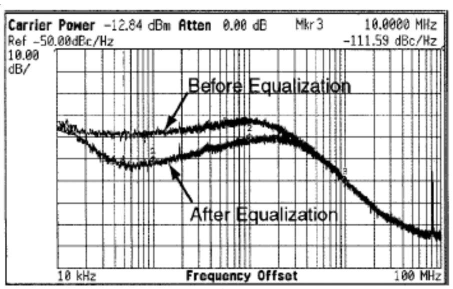


Fig. 7 Phase noise of the recovered clock with and without equalization.

Case Study: DDJ Equalizer

IEEE JOURNAL OF SOLID-STATE CIRCUITS, VOL. 41, NO. 3, MARCH 2006 607

Analysis and Equalization of Data-Dependent Jitter

James F. Buckwalter, *Student Member, IEEE,* and Ali Hajimiri, *Member, IEEE*

***Abstract*—Data-dependent jitter limits the bit-error rate (BER) performance of broadband communication systems and aggravates synchronization in phase-and delay-locked loops used for data recovery. A method for calculating the data-dependent jitter in broadband systems from the pulse response is discussed. The impact of jitter on conventional clock and data recovery circuits is studied in the time and frequency domain. The deterministic nature of data-dependent jitter suggests equalization techniques suitable for high-speed circuits. Two equalizer circuit implementations are presented. The first is a SiGe clock and data recovery circuit modified to incorporate a deterministic jitter equalizer. This circuit demonstrates the reduction of jitter in the recovered clock. The second circuit is a MOS implementation of a jitter equalizer with independent control of the rising and falling edge timing. This equalizer demonstrates improvement of the timing margins that achieve 10^{-12} BER from 30 to 52 ps at 10 Gb/s.**

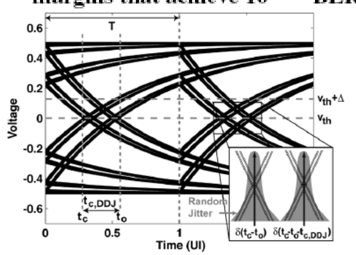


Fig. 3. Definition and characterization of the timing deviations in the eye.

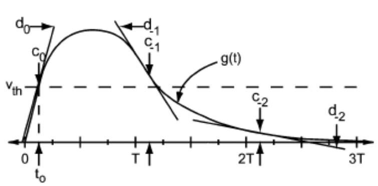


Fig. 4. Description of pulse response for DDJ equalization.

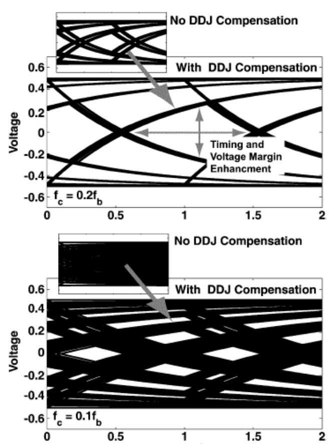


Fig. 9. Opening a closed data eye with the use of only DDJ compensation.

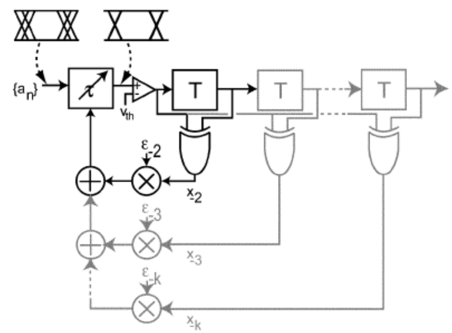


Fig. 7. Block diagram for the DJE and the extension, shown in gray, for compensating over multiple samples.

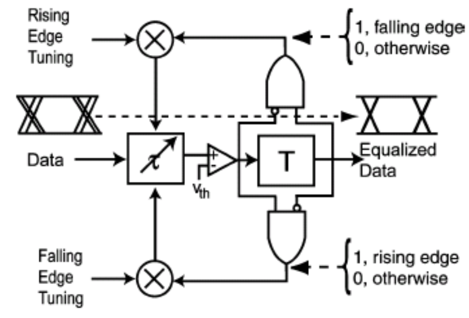


Fig. 8. DJE with independent control of the rising and falling edges.

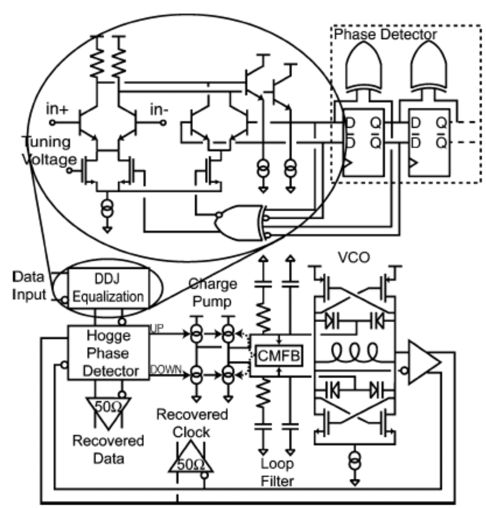


Fig. 12. Block diagram for the DJE CDR.

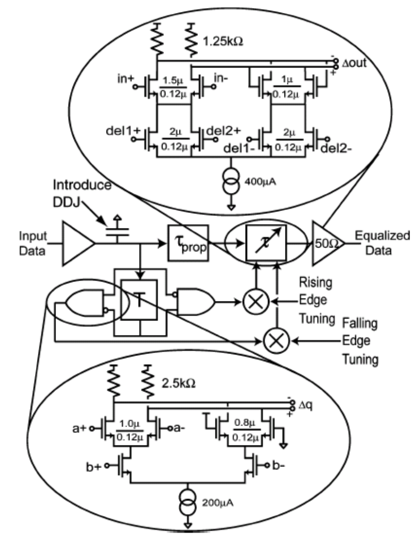


Fig. 14. Schematic for the CMOS DJE. Inset includes schematic for low-power CML AND gate and delay cell.

Case Study: Cross-Coupling Jitter Equalizer

IEEE JOURNAL OF SOLID-STATE CIRCUITS, VOL. 41, NO. 3, MARCH 2006

Cancellation of Crosstalk-Induced Jitter

James F. Buckwalter, *Student Member, IEEE*, and Ali Hajimiri, *Member, IEEE*

***Abstract*—A novel jitter equalization circuit is presented that addresses crosstalk-induced jitter in high-speed serial links. A simple model of electromagnetic coupling demonstrates the generation of crosstalk-induced jitter. The analysis highlights unique aspects of crosstalk-induced jitter that differ from far-end crosstalk. The model is used to predict the crosstalk-induced jitter in 2-PAM and 4-PAM, which is compared to measurement. Furthermore, the model suggests an equalizer that compensates for the data-induced electromagnetic coupling between adjacent links and is suitable for pre- or post-emphasis schemes. The circuits are implemented using 130-nm MOSFETs and operate at 5–10 Gb/s. The results demonstrate reduced deterministic jitter and lower bit-error rate (BER). At 10 Gb/s, the crosstalk-induced jitter equalizer opens the eye at 10^{-12} BER from 17 to 45 ps and lowers the rms jitter from 8.7 to 6.3 ps.**

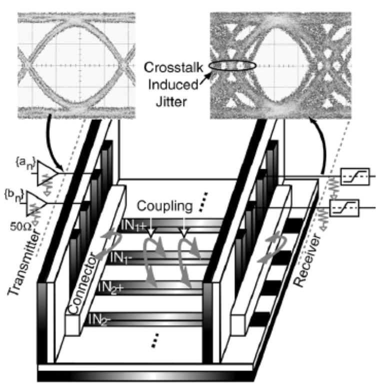


Fig. 1. Crosstalk jitter generated in the data eye due to the influence of a neighboring signal.

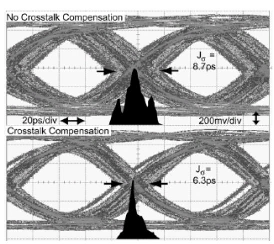


Fig. 16. Data eyes at 10 Gb/s before and after equalization.

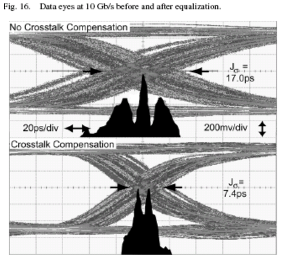


Fig. 17. Data eyes at 5 Gb/s before and after equalization.

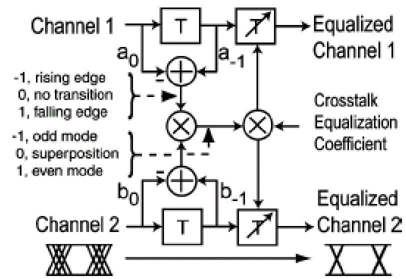


Fig. 9. Schematic of a two-channel crosstalk-induced jitter equalization.

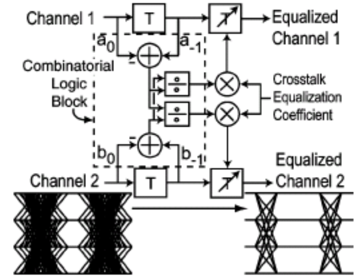


Fig. 10. Schematic of a two-channel 4-PAM crosstalk-induced jitter equalization.

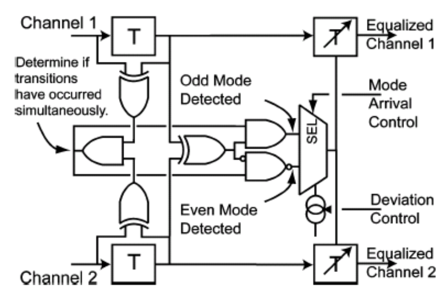


Fig. 12. Implemented version of two-channel CIJ equalizer.

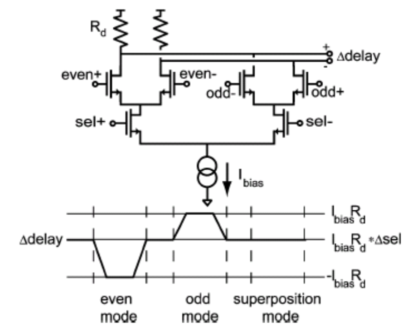


Fig. 13. Schematic and operation of the mode multiplexer.

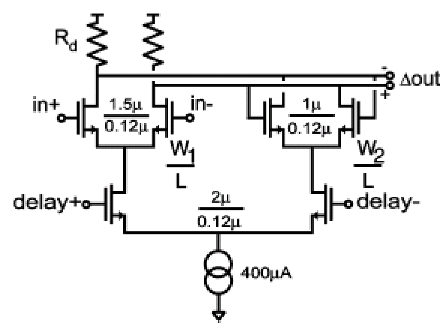
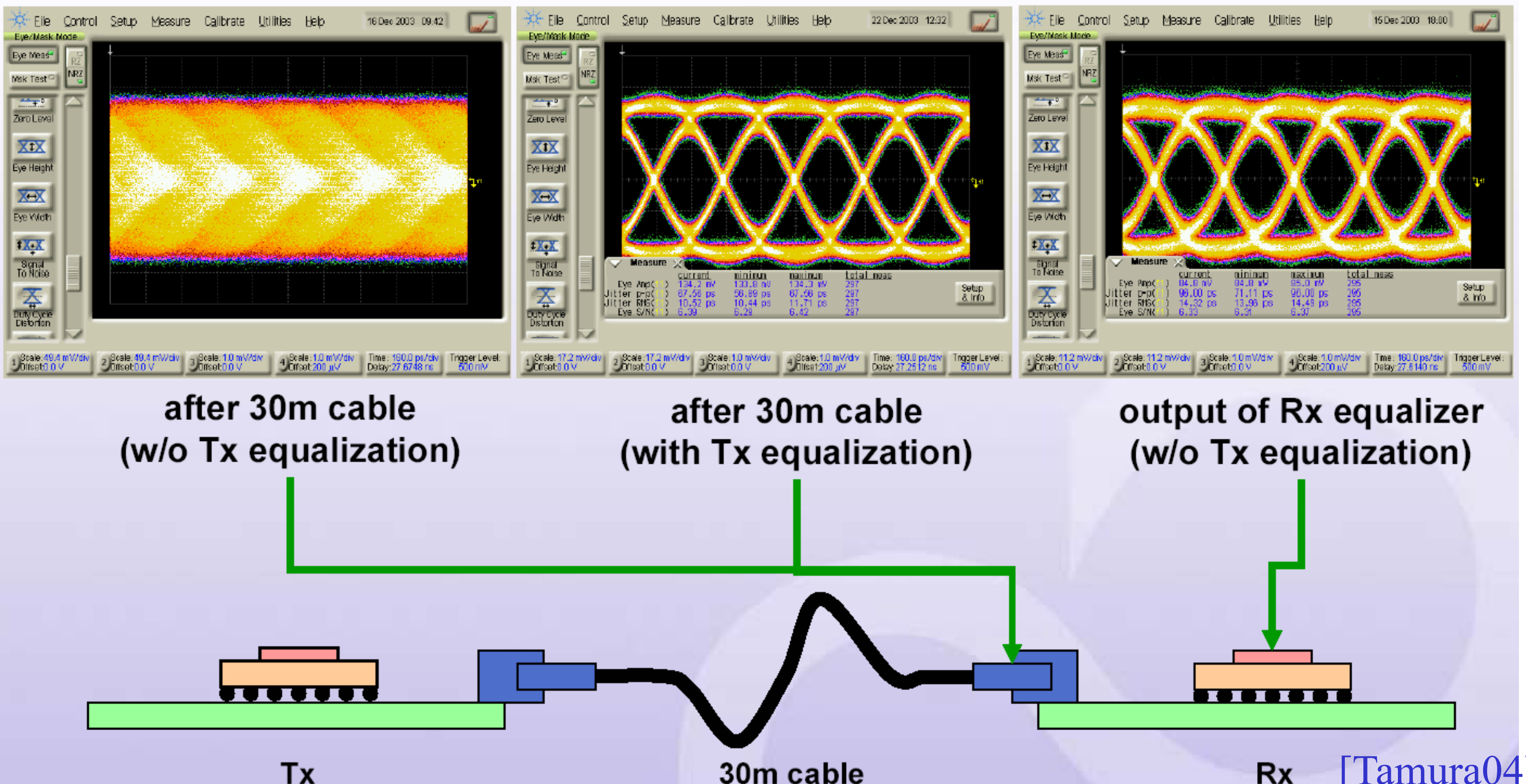


Fig. 14. Schematic of the delay cell.

Case Study: VLSI Equalizer Circuits

- **6.4Gb/s Transceiver with Tx FIR filter and Rx equalizer**



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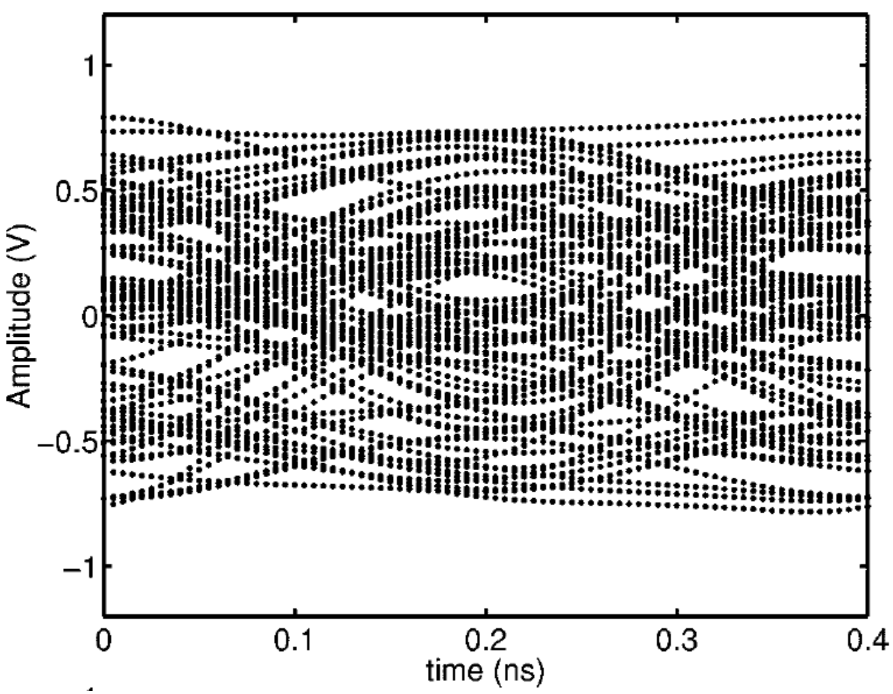
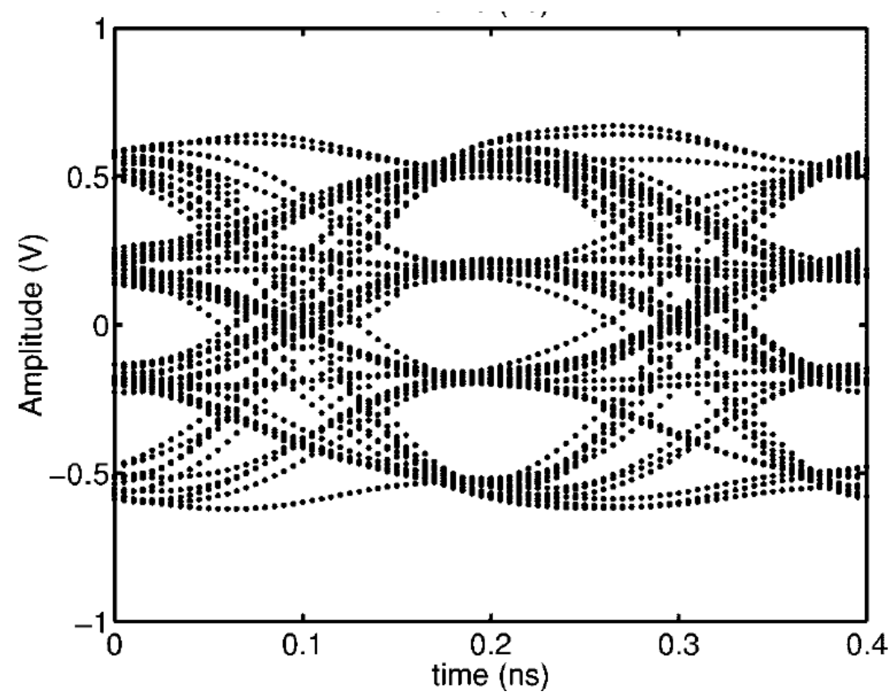
Case Study: Equalization Eye Diagram

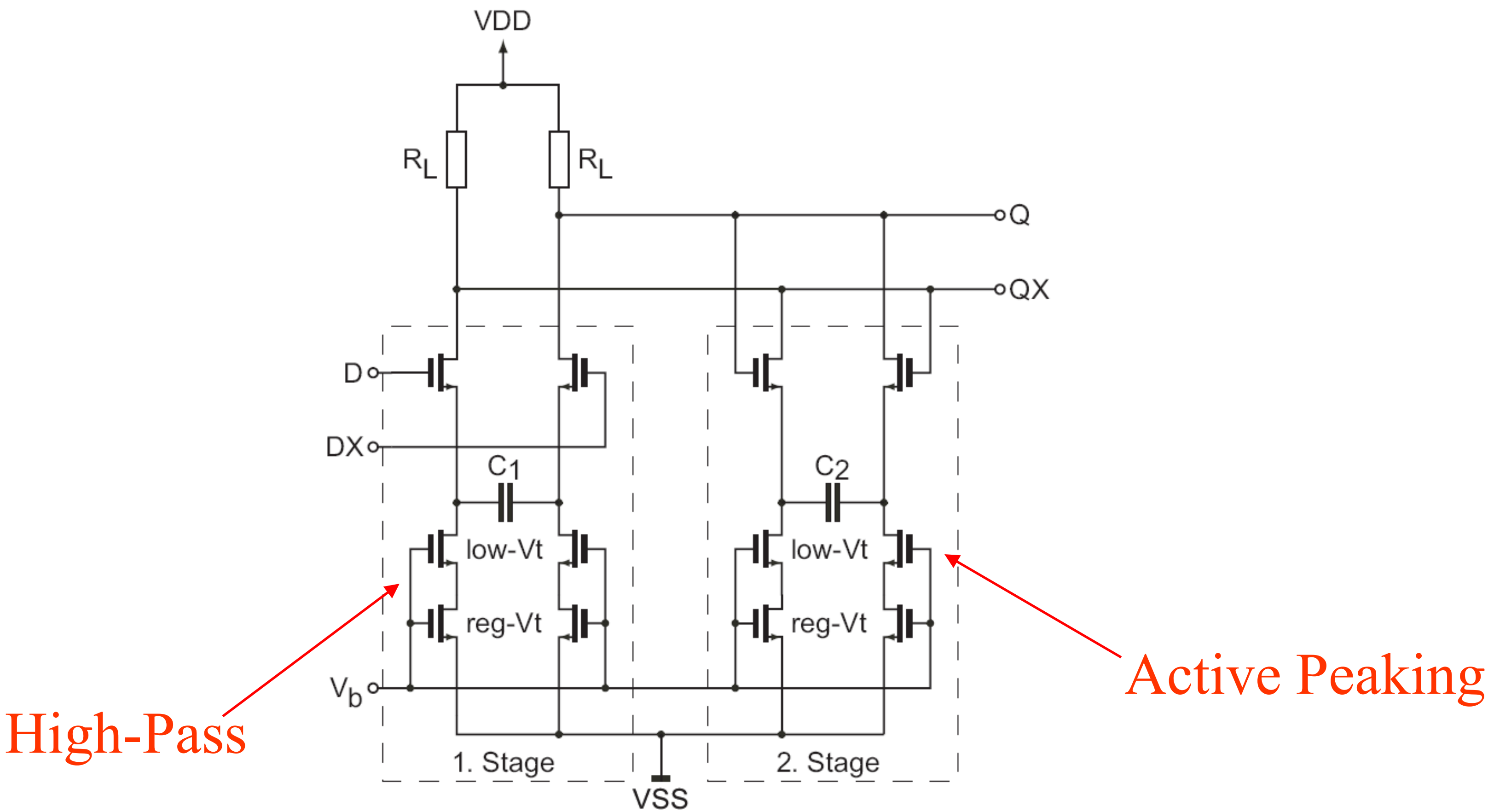


(a) Before equalization

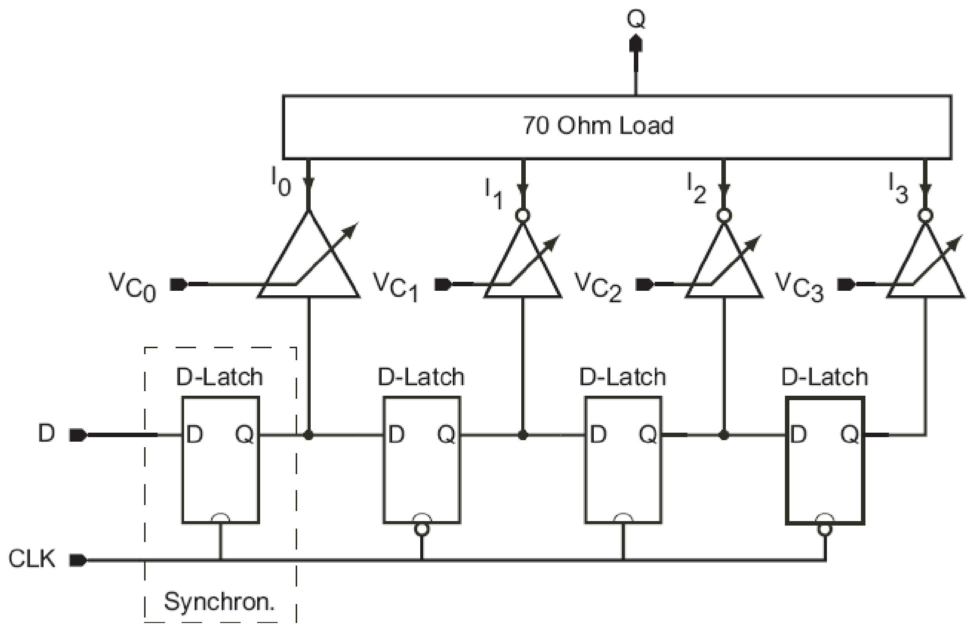


(b) After equalization

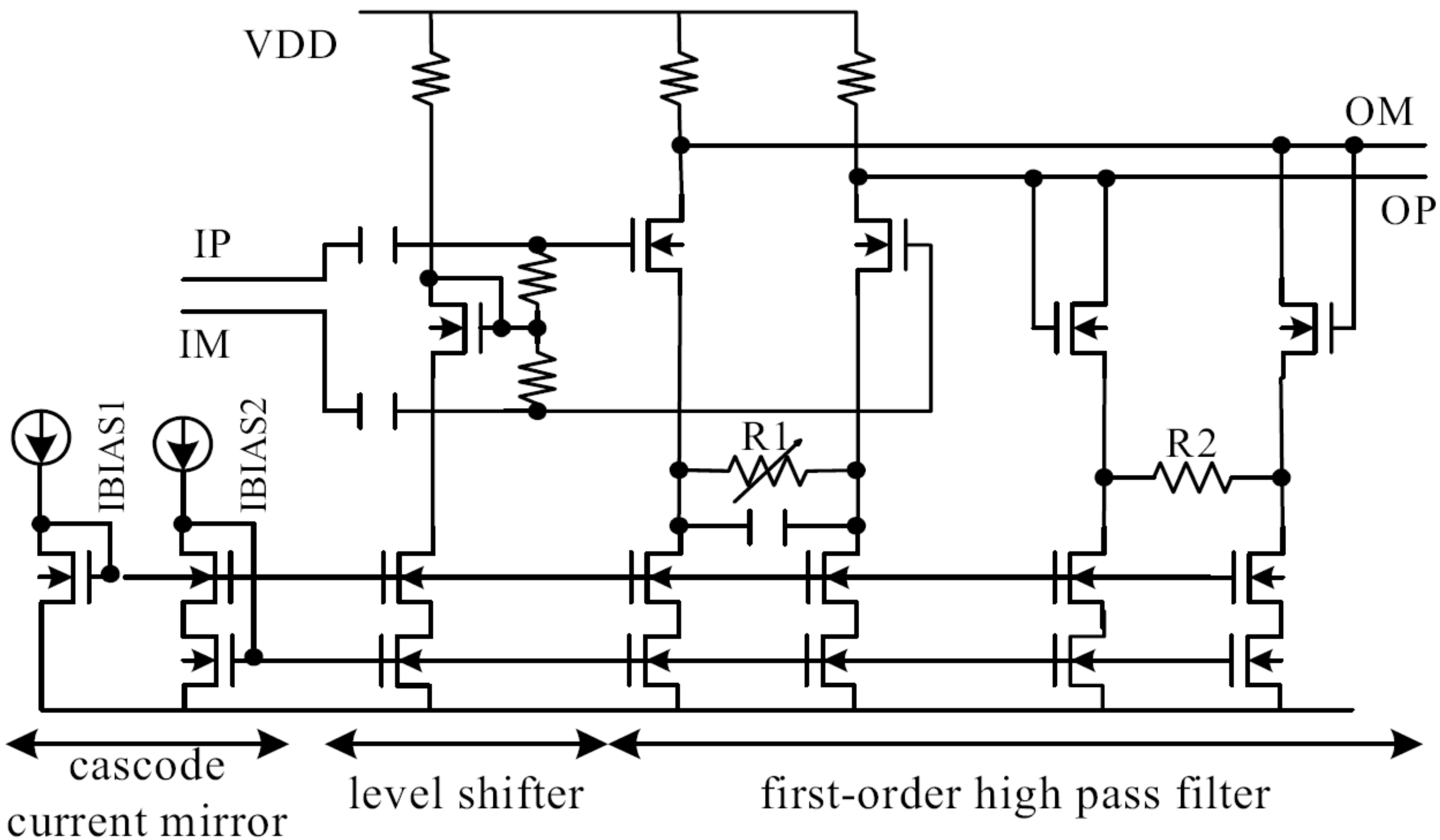
VLSI Equalization Circuit Implementations



VLSI Equalization Circuit Implementations



VLSI Equalization Circuit Implementations



Advanced Topic: Adaptive Equalization

- Adaptive Equalization techniques are typically used for
  - Unknown channel
  - Time-varying channel
- Several adaptive algorithms
  - Zero-Forcing (ZF)
    - Minimize the ISI at the sampler input
  - Mean square error (MSE)
    - Minimize the ISI + Noise at sampler input

Advanced Topic: Adaptive Equalization

- Several Mean Square Error (MSE) implementations
  - Recursive Mean Square (RMS) algorithm
    - For fast time-varying channel (fast training)
    - Complicated circuit implementation
    - Converge issue
  - Least Mean Square (LMS) algorithm
    - Implementation simplicity
    - Slower training (for slow time-varying channel)
  - Sign-Sign LMS (SS-LMS) algorithm
    - Further reduction in implementation simplicity
    - Even slower training (suitable for unknown or slow time-varying channel)